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Master's Thesis

**A Study of Large Language Models' Emotion
Classification in Ambiguous Text with Perceived
Authorship**

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Abstract

Emotion recognition in text is usually evaluated as though every text has one correct emotion label. In practice, many texts admit several plausible emotional readings, and the reading a person arrives at depends not only on the words but on what they assume about the person who wrote them. The sentence “*I can’t believe I actually did that*” can express pride, embarrassment, or regret depending on who is taken to have said it. This thesis asks whether perceived author identity measurably changes emotion interpretation, for human readers and for large language models, and whether the two respond in comparable ways.

Three hundred short self-reported emotional experiences were drawn from the crowd-enVent corpus and stratified into three categories of one hundred texts each: *unambiguous control texts*, *author-independent ambiguous texts*, and *author-relevant ambiguous texts*. Every text was presented under a neutral framing and under two contrasting author personas, with the text itself held constant. Human judgments were collected for all three framings, and seven large language models classified every text under four framings and six prompt configurations, yielding 7,200 model-ensemble distributions and 900 human distributions (one per text and framing condition), constructed from up to 50,400 model answers and 2,899 human ratings.

The thesis first tests its hypotheses by selecting one majority or top emotion per condition. It then repeats the tests with **full probability distributions over eleven emotions**. The comparison shows what the single-label approach cannot measure. Among the 300 sampled texts, 19.7% have no emotion commanding a strict majority of readers. Within that group, 6.3% still have a unique most-frequent (plurality) label, so a pipeline that reports one label per text can retain it while concealing how divided readers actually were; the remaining 13.3% have no unique top choice at all, forcing an arbitrary tie-break. Representing each judgment as a distribution preserves the disagreement instead of discarding it, and makes it possible to ask not merely *whether* an answer changed but *how far* and *in which direction* it moved.

Four hypotheses were tested. Human judgments shift significantly under author framing for author-relevant ambiguous texts and show no detectable shift for unambiguous controls, which is the predicted pattern; in the author-independent category only one persona contrast reaches significance, without evidence that the two persona effects differ. The model ensemble shifts significantly under both personas. A previously unanalysed **generic_human** control separates two comparisons under matched models: adding an unspecified author line alone moves classifications by about 8% on both sides, and moving on to a specific persona is associated with a separately measured 16–19% shift (the two distances are not additive components of one total effect). The predicted ordering across categories holds in raw shift magnitude, but a sensitivity analysis referencing each text against its own conditional sampling null shows that the ordering is not separable from baseline vote dispersion at this sample size. Finally, models track the *direction* of human shifts better than chance but only weakly, while human shifts are substantially larger in *magnitude* than model shifts; this magnitude difference survives standardizing both sides to an equal number of votes.

Two results are reported as negative or near-negative findings rather than smoothed away: almost no individual per-emotion shift survives multiple-comparison correction once both human and model shifts are tested (7 of 66, forming a small but directionally coherent model-side pattern rather than scattered noise), and no prompt-design effect on the size of the model framing effect was detected in the exploratory analysis. A further finding is methodological, and it is the sharpest illustration of why the representation matters: repeating the identical statistical test on the selected label rather than the full distribution rejects the null in **one** of the six tests that establish H1 under a reverse-alphabetical tie-break convention, and in **none** of the six under the alphabetical convention used throughout this thesis (only these two otherwise-arbitrary conventions were tested), where the distributional treatment rejects it in three regardless. With a selected label, this study would have detected a statistically significant human framing effect in one of its six H1 tests at best, or none under this thesis’s own convention. More broadly, across all comparisons, roughly two fifths of both human and model comparisons moved in ways a selected-label analysis would have recorded as no change at all; this too is convention-dependent, since crediting the label method for tracking the complete set of tied top emotions rather than only the alphabetically selected one lowers the human figure to about a quarter (25.8%), leaving the model figure materially unchanged (38.7%).

Taken together, the results support the view that author framing is a real and measurable influence on emotion interpretation, that language models are sensitive to it, and that the resemblance between human and model behaviour is partial and directional rather than close.

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List of Abbreviations

ρ	Spearman’s rank correlation coefficient
κ	Fleiss’ kappa, the inter-annotator agreement statistic used in Section 5.19
AI	Artificial Intelligence
API	Application Programming Interface
BH-FDR	Benjamini–Hochberg False Discovery Rate correction; used for exploratory test families
CI	Confidence Interval
Clustered bootstrap	Bootstrap resampling texts rather than rows, so that repeated observations of the same text are not treated as independent
Cosine similarity	Angle-based similarity between two delta vectors, used to compare shift direction
crowd-enVent	Corpus of self-reported emotional events with author and reader annotations (Troiano et al., 2023)
Delta vector	Signed difference between a persona-condition vector and its neutral-condition vector
Dominant set	The set of emotions attaining the maximum probability in a vector; a set rather than a single label because ties are frequent
Excess TV	Observed TV minus the TV expected for that text under its own conditional sampling null
Few-shot	Prompt variant providing several worked examples
Generic human	Framing condition attributing the text to an unspecified human author
H1–H4	The four research hypotheses stated in Section 3.2
Holm	Holm correction for multiple comparisons, controlling the family-wise error rate; used for confirmatory tests

LLM	Large Language Model
Neutral	Framing condition in which no author information is supplied
Numbered inline	/ The two label-presentation formats: emotions as a numbered list, or embedded in the question text
One-shot	Prompt variant providing a single worked example
Paired sign-flip	Permutation null in which each model's own two answers are swapped with probability one half, holding model identity fixed
Permutation test	Significance test comparing an observed statistic against a null built by relabelling the observed data
Persona A / B	The two contrasting author personas assigned to each text
pp	Percentage points
Pseudoreplication	Treating repeated, correlated observations of the same unit as independent observations
QC	Quality Control
RQ	Research Question
Support size	Number of emotions receiving non-zero probability mass
TV	Total Variation distance, $\frac{1}{2} \sum_i p_i - q_i $; the share of probability mass separating two distributions
Zero-shot	Prompt variant providing no worked example

1 Introduction

Emotion recognition in text aims to identify the affective state expressed or implied in written language. Unlike sentiment analysis, which typically reduces to positive and negative polarity, emotion classification assigns finer-grained categories such as joy, anger, fear, shame, pride or sadness. The task is difficult because emotional meaning is rarely encoded explicitly. It emerges from an interaction between linguistic content, situational context, and the assumptions a reader makes about the person speaking.

A central challenge is ambiguity. Many texts permit several plausible emotional readings, whether because context is underspecified, because the situation is emotionally complex, or because writer and reader occupy different perspectives. Even short and apparently simple statements can be read in competing ways. Consider:

I can't believe I actually did that.

With no further context this could express pride, embarrassment, regret, excitement or disbelief. Readers typically resolve such ambiguity by drawing on background assumptions, and one of the most readily available assumptions concerns *who* is speaking. The same sentence attributed to a first-time marathon finisher and to someone recounting a lapse of judgment invites two different emotional readings, despite the words being identical.

1.1 Motivation

The motivation for this thesis is twofold.

From a computational perspective, large language models are increasingly deployed for text classification tasks including emotion recognition, and they are known to be sensitive to prompt framing and contextual cues. If minimal author information systematically shifts a model's emotional interpretation, then an emotion classifier's output depends

on social context supplied in the prompt, not on text content alone. That has direct consequences for how such systems are evaluated and deployed.

From a cognitive and social perspective, the question is whether models mirror human interpretive behaviour when handling ambiguous emotional content. Humans demonstrably use assumptions about a speaker to disambiguate. Whether models reproduce that behaviour, and whether they reproduce it in the same direction and to the same degree, is an empirical question that has not been directly tested by manipulating perceived authorship while holding text constant.

1.2 Problem Statement

Existing work on emotion recognition treats disagreement between annotators primarily as noise to be resolved. Datasets are typically distributed with one aggregated label per text, most often by majority vote, and models are trained and evaluated against that label. This practice embeds an assumption that is false for a substantial share of ambiguous texts: that a single correct emotional reading exists and that annotators who deviate from it are simply wrong.

The problem this thesis addresses has two parts. First, whether perceived author identity is a systematic influence on emotion interpretation rather than an incidental one, in humans and in models. Second, whether the standard practice of collapsing judgments to a single label is capable of detecting such an influence at all. If author framing shifts the *distribution* of plausible readings without always changing which reading is most popular, then a majority-label analysis will record no change where a real change occurred.

1.3 Research Approach

This thesis manipulates perceived authorship experimentally while holding textual content fixed. Three hundred texts from the crowd-enVent corpus [Troiano, Oberländer, and Klinger 2023] were stratified into three categories: unambiguous control texts, author-independent ambiguous texts, and author-relevant ambiguous texts. Each text was then presented under a neutral framing and under two contrasting author personas. Human

annotations were collected for all framings, and seven large language models classified every text across four framings and six prompt configurations.

The analytical commitment that distinguishes this work is representational. Every judgment, human or model alike, is represented as a full probability distribution over the eleven emotions in the label set, rather than as a single label. Shift is then measured as movement of that distribution: its *magnitude* by total variation distance, and its *direction* by the signed change in probability mass across emotions. This makes it possible to distinguish a text whose interpretation barely moved from one that was substantially reinterpreted but happened to retain the same most-popular answer.

1.4 Contributions

This thesis makes the following contributions.

- An experimental design that isolates the effect of perceived authorship on emotion interpretation by manipulating author framing while holding text constant, applied in parallel to human annotators and to seven large language models.
- A distributional analysis framework in which every judgment is an eleven-dimensional probability vector, with magnitude and direction of framing-induced shift measured separately and never conflated.
- Empirical evidence that author framing measurably shifts both human and model emotion judgments, together with a control condition isolating the effect of the persona’s content from the mere presence of an author attribution.
- A direct quantification of what majority-label analysis discards: the proportion of judgments that moved but would have been recorded as unchanged.
- An explicit treatment of what these data can and cannot establish, including a sensitivity analysis showing that the predicted ordering across ambiguity categories is not separable from baseline disagreement at this sample size.

1.5 Thesis Structure

Chapter 2 reviews work on annotator disagreement in emotion recognition and on framing effects arising from perceived authorship, and identifies the gap this thesis addresses.

Chapter 3 states the research questions and hypotheses and describes the corpus, the categorisation of texts by ambiguity type, and the author-framing manipulation. Chapter 4 documents the full pipeline: prompt construction, model evaluation, the human annotation study, the construction of emotion probability vectors, the metrics derived from them, and the quality-control and statistical procedures. Chapter 5 reports the results, organised by hypothesis. Chapter 6 interprets these findings against the research questions, sets out the limitations of the design, and identifies directions for further work. Chapter 7 concludes.

2 Theoretical Background

This chapter reviews the two research strands this thesis brings together. The first concerns emotion recognition and the status of annotator disagreement within it. The second concerns framing effects arising from perceived authorship. Section 2.3 identifies where the two meet, and Section 2.4 states the gap this thesis addresses.

2.1 Emotion Recognition and Annotator Disagreement

Emotion recognition assigns fine-grained affective categories to text, in contrast to sentiment analysis, which typically reduces to polarity. Widely used resources such as GoEmotions [Demszky et al. 2020] provide large-scale annotated corpora over rich emotion taxonomies, and transformer-based classifiers trained on them achieve strong benchmark performance [Acheampong, Nunoo-Mensah, and Chen 2021].

These benchmarks share a structural assumption. Multiple annotators label each text, their judgments are aggregated (most often by majority vote), and the aggregate is treated as the gold standard against which models are trained and scored [Basile 2020]. The assumption is that annotators are interchangeable estimators of one underlying truth, and that deviation from the majority is error.

A substantial body of work argues this assumption fails for subjective tasks. Pavlick and Kwiatkowski 2019 show that disagreement in natural language inference is frequently *inherent* rather than erroneous: annotators genuinely and reproducibly differ. Uma et al. 2021 survey learning approaches that model disagreement rather than eliminate it. Aroyo and Welty 2015 argue directly against the single-ground-truth framing, and Davani et al. 2022 show that modelling annotators individually can outperform training on aggregated labels.

Where disagreement is systematic, aggregation is not neutral. Annotator judgments have been shown to vary with demographic background [Prabhakaran, Davani, and

Díaz 2021], and in hate speech and offensive language annotation this variation tracks social group membership [Sap et al. 2019; Waseem 2016]. The variation is not only demographic: Geva et al. 2019 show, for datasets whose examples were themselves written by crowdworkers, that a model can learn to recognise the most prolific annotators from the text they authored, and that it generalises poorly to examples written by annotators unseen in training, evidence that annotator identity itself carries systematic, non-interchangeable signal. Collapsing to a majority label therefore encodes the dominant reading and suppresses minority ones [Davani, Díaz, and Prabhakaran 2022].

Emotion is a particularly clear case. Acheampong et al. 2021 note that emotional meaning is often implicit and context-dependent, which leaves room for competing but defensible readings. Chou and Lee 2019 argue that every rating carries information and that discarding non-majority ratings discards evidence.

2.2 Perceived Authorship and Framing Effects

The second strand concerns how assumptions about a speaker shape interpretation of what is said. Identical text can be read differently depending on who is believed to have produced it.

Henestrosa et al. 2023 found that, for a science journalism article held constant in content, authorship cues affected readers’ perceptions of the *author*, while their studies detected no differences in the article’s perceived credibility or trustworthiness between AI and human authorship conditions. The manipulation is minimal, just an attribution line rather than a change in content, and yet it moves some judgments and not others. This is the effect structure the present thesis adopts: hold the text fixed, vary only the attributed author, and measure the change, without assuming in advance which judgments will and will not respond to it.

For language models, sensitivity to contextual framing is well documented. Santurkar et al. 2023 show that model outputs on opinion questions align with some demographic groups more than others, and that this alignment shifts when the model is prompted to adopt a persona. Zhang et al. 2018 show that conditioning dialogue agents on persona profiles systematically changes their responses. Both establish that persona information in a prompt is not inert.

Work applying models to ambiguous emotional content has begun to treat ambiguity as a first-class property. Hong et al. 2025 propose ambiguity-aware emotion recognition with large language models, acknowledging that a single label is not always the right output target. Röttger et al. 2022 distinguish annotation paradigms that seek one correct answer from those that deliberately capture variation.

2.3 Points of Intersection

The two strands intersect in a way neither has fully exploited.

The disagreement literature establishes that variation in subjective annotation is systematic and worth preserving, but it generally treats that variation as a property of the *annotator*: their background, their sensitivity, their interpretive style. The variable is who is reading.

The framing literature establishes that assumptions about the *author* shift interpretation, but typically measures the shift on a single outcome variable such as perceived quality, and typically in humans or in models, not both under one design.

Combining them suggests a question neither asks directly. If disagreement in emotion annotation is partly driven by differing assumptions about the author, assumptions readers make silently when context is underspecified, then supplying that context explicitly should move the *distribution* of emotional readings in a measurable and predictable way. And if models are sensitive to persona framing, they should exhibit an analogous shift, which can be compared against the human one on the same texts.

2.4 Research Gap and Motivation

Three gaps follow from the review above.

Perceived authorship is not manipulated as an experimental variable in emotion recognition. Existing emotion corpora record disagreement but do not vary author attribution while holding text constant. Disagreement is observed, not induced, so its relationship to assumptions about the author cannot be isolated.

Aggregation destroys the quantity of interest. If author framing redistributes probability mass across plausible emotions without always changing which emotion is

most popular, then a majority-label analysis is structurally incapable of detecting it. The effect can only be observed if judgments are represented as distributions. This thesis quantifies directly how much is lost by aggregation (Section 5.6).

Human and model framing effects are rarely measured under one design.

Claims that models resemble, or fail to resemble, human interpretation usually compare against a majority-derived gold label, which answers whether the model’s single answer matches the human’s single answer. It does not answer whether the model *responds to context* the way humans do. Measuring both sides’ shift under the same manipulation, on the same texts, and comparing direction and magnitude separately, is a different and more targeted question.

This thesis addresses all three. It manipulates perceived authorship experimentally, represents every judgment as a full emotion distribution, and applies the identical manipulation to human annotators and to seven large language models so that their responses can be compared directly.

3 Methodology

This chapter states the research questions and hypotheses, then describes the corpus, the categorisation of texts by ambiguity type, and the author-framing manipulation. It closes with an overview of the realised experimental design and an explicit account of where that design departs from the approved proposal.

3.1 Research Questions

The thesis addresses three research questions.

- **RQ1 (Framing Effect in Humans).** Does perceived author identity significantly influence human emotion judgments of ambiguous texts?
- **RQ2 (Framing Effect in LLMs).** Does perceived author identity significantly influence emotion classifications produced by large language models?
- **RQ3 (Human–Model Alignment).** Do LLMs exhibit framing effects that are similar in direction and magnitude to those observed in human annotators?

3.2 Research Hypotheses

Four hypotheses operationalise these questions.

- **H1 (Human Framing Hypothesis).** Human emotion judgments will differ significantly across author-framing conditions for ambiguous texts, but not (or substantially less) for unambiguous control texts.
- **H2 (Model Framing Hypothesis).** LLM emotion classifications will shift across author-framing conditions, indicating sensitivity to minimal author cues.

- **H3 (Ambiguity Sensitivity Hypothesis).** Framing effects will be stronger for author-relevant ambiguous texts than for author-independent ambiguous texts.
- **H4 (Alignment Hypothesis).** The direction of framing-induced shifts in LLM classifications will partially align with human judgment shifts, but the magnitude of these shifts will differ.

Two points about H4 require emphasis, because they determine how its evidence is scored. H4 predicts *partial* directional alignment, so a positive but modest directional correlation is evidence *for* the hypothesis, not against it. H4 also predicts that magnitudes will *differ*; finding human and model shift sizes to be similar would therefore be a surprise relative to the hypothesis as stated, not a confirmation of it. The two components are scored separately throughout Chapter 5.

No numbered hypothesis concerning prompt structure exists in the approved proposal. Analysis of prompt-configuration effects is accordingly treated as exploratory throughout and is never presented as a test of a formal hypothesis.

3.3 Corpus Selection and Dataset Construction

The data source is the crowd-enVent corpus [Troiano, Oberländer, and Klinger 2023], which consists of short self-authored descriptions of emotional events. Each text carries two annotation layers: a label supplied by the original author, stating the emotion they intended to convey, and labels supplied by five independent readers who did not see the author’s label. This dual-layer structure is what makes the corpus suitable here, because it renders disagreement measurable at two distinct levels: between author and readers, and among readers themselves.

The corpus uses eleven emotion categories: *anger*, *disgust*, *fear*, *guilt*, *joy*, *pride*, *relief*, *sadness*, *shame*, *surprise* and *trust*. All eleven are retained. The set is small enough to remain manageable for human annotators and for models, and excluding any category would risk suppressing precisely the disagreement patterns the study is designed to observe. This fixed, alphabetically ordered set of eleven is used as the dimension set for every probability vector constructed in Chapter 4.

Ambiguity was operationalised through two quantities computed per text. **Author–reader agreement** asks whether the reader majority recovers the emotion the author states they intended; a mismatch means the text did not communicate what its author

meant. **Reader consensus** (*majority share*) is the fraction of the five readers choosing the majority emotion. Because there are exactly five readers, this measure takes only the values 0.2, 0.4, 0.6, 0.8 and 1.0, a coarseness that directly constrains where a threshold can sit.

Reader consensus is not concentrated at the top of that scale. A majority share of 0.6 (three readers of five) is the single most common outcome in the candidate corpus. Disagreement is the normal case, not an edge case, which is the empirical basis for the distributional representation adopted in Section 4.3.

3.3.1 The classification rule

Three decisions are applied in a fixed order. Their thresholds and ordering are encoded in `scripts/classify_ambiguity_pools.py`. **The persona-flip input for each text (Step 2 below) was generated by an LLM judge, not by human pilot annotators:** the judge was shown the same text under two candidate personas and asked for the emotion it read into each, and the two judgments were compared directly. This LLM-judging step is described in the dataset-construction notebook, `complete_analysis.ipynb`; the specific model used as the judge is not recorded in the materials retained for this thesis and could not be reconstructed at the time of writing. The full derivation, including a threshold sensitivity analysis, is documented in the dataset-construction notebook `00_corpus_and_sampling.ipynb`.

Step 0: exclude off-label texts. Texts whose author emotion or reader majority is *boredom* or *no-emotion* are dropped, as these fall outside the eleven-emotion set. Forcing them into the taxonomy produces persona-to-persona differences that are a labelling artefact rather than a real interpretive shift. This removes 215 of the 1,200 candidates, leaving 985 eligible.

Step 1: threshold first. A text is **unambiguous** if majority share ≥ 0.8 *and* the readers recover the author’s own label. This is decided purely from the original corpus’s reader data.

Step 2: flip test, among the remainder only. Of the texts not classified as unambiguous, those whose two candidate personas yield different judged emotions from the LLM judge are **author-relevant ambiguous**; the rest are **author-independent ambiguous**.

The ordering is deliberate and matters. A persona flip does *not* override strong, author-matching reader consensus: a text readers already agree on remains unambiguous even if a persona pair can be found that moves the LLM judge. Reversing the order would allow the flip test to recruit high-consensus texts into the author-relevant group, making that category’s definition circular with respect to H3.

A filtering inconsistency in this step. Step 0 excludes texts whose author emotion or reader majority falls outside the eleven-emotion set, on the grounds that forcing an off-label judgment into the taxonomy is a labelling artefact rather than a real interpretive shift. The same standard was not applied to the LLM judge’s own answers at Step 2: of the 100 texts classified as author-relevant, 24 have the judge answering *no-emotion* for at least one persona, which is exactly the kind of off-label judgment Step 0 treats as unreliable elsewhere. These texts were not re-screened against that standard, so the author-relevant category is not filtered as consistently as the unambiguous category is.

The threshold of 0.8 is one of only three defensible settings given the five-valued measure. At 0.6 the unambiguous pool would absorb texts on which only three of five readers agreed, weakening the control category the design depends on; at 1.0 it would admit only unanimous texts. All three settings leave more than 100 candidates per pool, so 0.8 is not forced by availability; it is chosen as the stricter of the two middle options.

3.3.2 Drawing the sample

The sample was not produced by a single fresh draw, and describing it as one would misstate what is reproducible. The procedure is a **reconciliation** against an existing sample: the current sample is retained, every text in it is reclassified under the rule above, any pool holding more than 100 texts is trimmed back to 100 with the fixed seed (42), and shortfalls are replaced by drawing from eligible candidates not already used. Three texts reserved as one-shot and few-shot prompt examples are excluded throughout, so that no sampled text can appear inside its own prompt. The result is asserted to contain 300 texts, 100 per pool, with no duplicates.

Sampling fractions differ across pools because the pools differ in size; the design prioritises equal n per category over proportional representation, so that category comparisons are not confounded with unequal sample size.

What is and is not reproducible. The classification rule is fully reproducible and is verified against all 300 sampled texts. The *specific membership* of the sample is not

recoverable from the seed alone, because it depends on the prior sample’s contents and on the candidate ordering at the time each replacement was drawn. `sampled_texts.csv` is therefore the authoritative record of which texts were used.

Seven verification checks on the sample (size, balance, absence of reserved examples, absence of duplicates, pool eligibility, agreement with the classification rule, and absence of off-label texts) all pass, and are enforced as assertions rather than advisory output in `00_corpus_and_sampling.ipynb`.

One limit of the off-label filter. Step 0 screens two labels per text, the author’s own and the selected reader-majority label, not the individual reader votes. A text can therefore enter the sample while still containing reader votes outside the eleven-emotion set, and 27 such votes survive across 25 texts, so **275 of the 300 sampled texts retain all five in-inventory reader votes**. Those votes are excluded when the original-corpus vectors are built, which is why vector construction tracks n per vector rather than assuming five throughout (Section 4.3.1).

3.4 Ambiguity Types and Author Relevance

Not all ambiguous texts are ambiguous for the same reason. In some, ambiguity arises because author-related context is missing; in others it stems from an underspecified or emotionally complex situation in which the author’s identity is immaterial. The distinction matters directly for H3, which predicts that framing effects concentrate in the former.

Texts were therefore categorised into three groups, which are used consistently throughout this thesis:

- **Author-relevant ambiguous texts**, where knowledge of the author’s role, perspective or social position could plausibly shift the perceived emotion. A description of a risky action may read as pride from the actor and as fear or concern from an observer.
- **Author-independent ambiguous texts**, where several emotions remain plausible regardless of who wrote the text, typically because situational detail is missing rather than because author identity is at issue.

- **Unambiguous control texts**, drawn from the high-agreement pool, where author and readers largely converge on one emotion.

The final sample comprises **300 texts, balanced at exactly 100 per category**. This balance is a property of the sampling design and is verified programmatically as part of the quality-control gate described in Section 4.5.

A property of this categorisation must be stated plainly because it constrains what H3 can claim. Texts were assigned to the author-relevant group partly on the basis of an LLM judge’s own sensitivity to whether author framing would flip the reading: the corpus preparation records, for each text, two candidate persona readings from that judge and whether they diverge. Texts were therefore *selected* into that group partly on the property H3 subsequently tests. H3 is consequently a **transfer comparison**: it asks whether a grouping derived from one LLM’s judgments predicts framing effects in newly collected human data. It is not an independent discovery that author-relevance is the operative principle, and it is not worded as one in Chapter 5.

This has a sharper consequence for the model-side ordering than for the human side. Because the category itself was defined by whether an LLM’s judgment flips between personas, the finding that the *model ensemble’s* framing-effect magnitude is ordered author-relevant > author-independent > unambiguous (Section 5.9.1) is partly true by construction: texts were selected into the author-relevant pool precisely because at least one LLM’s reading was sensitive to the persona pair. That the ordering holds across all six prompt configurations, and for an ensemble of seven different models rather than only the one used as the classification judge, is evidence the pattern is not solely an artefact of the selection; but the model-side ordering claim should be read as partially circular rather than as an independent confirmation, in the same way the human-side H3 result is already treated as a transfer comparison rather than a fresh discovery.

3.5 Author-Framing Manipulation

The independent variable is perceived authorship. For every text, the textual content is held *exactly* constant and only a short author cue attached to it is varied, so any difference in emotion judgment cannot come from a change in the text itself. The cue is appended after the text, not prepended before it (Section 4.1.1 gives the exact format); which side of the text the cue sits on is not expected to matter for the manipulation’s logic, but the

earlier description of it as *preceding* the text did not match what was actually presented. This does not by itself isolate identity from every other kind of information the cue carries: a persona description can also supply substantive situational detail beyond who the author is (for instance, a persona pair contrasting someone who has struggled all year with someone expecting to pass easily differs in circumstance as well as in identity), so the design measures sensitivity to the supplied author/persona framing as a whole, not a pure identity mechanism in isolation.

Four framing conditions were used:

- **Neutral:** no author information is supplied.
- **Generic human:** the text is attributed to an unspecified human author.
- **Persona A** and **Persona B:** the text is attributed to a person with a defined social role.

The generic human condition is analytically important and is not merely an additional level. The neutral and persona prompts differ in more than persona content: the persona prompts add an author line that the neutral prompts lack entirely. A neutral-to-persona comparison therefore confounds *who the author is* with *whether an author is named at all*. The generic human condition separates the two, and Section 5.10 uses it for exactly that purpose.

Persona selection required care. A single fixed pair such as *politician* and *comedian* is not equally relevant across texts; a domestic anecdote has no natural connection to either role. Personas were therefore assigned per text to fit its topic, with the two personas of a pair chosen to represent opposing interpretive expectations (one associated with accountability and one with latitude, for instance) so that a framing effect, if present, has a reasonable opportunity to surface. Both personas for every text are recorded explicitly in the dataset alongside the rationale for the pairing.

3.6 Realised Experimental Design

Table 3.1 summarises the design as executed.

Human annotation. Participants were recruited and presented with texts one at a time together with the assigned author framing, selecting a label from the same fixed set of eleven emotions used for the models. Texts and framings were distributed such that

Table 3.1: Realised experimental design.

Factor	Levels
Texts	balanced across 3 ambiguity categories
Emotion labels	fixed, alphabetically ordered
Framing (models)	neutral, generic human, persona A, persona B
Framing (humans)	neutral, persona A, persona B
Models	7 (proprietary, commercial and open-weight)
Prompt variants	zero-shot, one-shot, few-shot
Label formats	numbered list, inline
Model judgments	$300 \times 4 \times 3 \times 2 = 7,200$ cells, each an ensemble vector aggregating up to 7 models' raw answers (50%)
Human judgments	300×3 sides

each text–framing pair received multiple independent annotations while **no annotator encountered the same text more than once**. This is a design property specified in the approved proposal, and it is the property on which the statistical treatment of the human data in Section 4.6 depends: because no participant saw more than one framing of a given text, the conditions for a text are rated by disjoint sets of people, and there is no within-text pairing across conditions to preserve.

Model evaluation. Each model saw each text exactly once per framing condition and prompt configuration, avoiding contamination across repeated exposure. For every (text, framing, variant, format, model) combination the predicted label was recorded. Prompt configurations are never pooled with one another in any analysis; they are reported separately or averaged only after being computed independently.

3.7 Deviations from the Approved Proposal

The following departures from the proposal are recorded here rather than left implicit.

Sample size. The proposal anticipated 300–400 texts. The realised sample is 300, chosen to permit exact balance at 100 per category.

Number of models. The proposal specified at least one proprietary, one commercial and one open-source model. Seven were used, which materially strengthens RQ2: a framing effect observed across seven independently developed systems is less readily attributed to an idiosyncrasy of one model family.

Number of prompt configurations. The proposal specified at least two prompt variants. Six configurations were used (three variants crossed with two label formats), allowing prompt sensitivity to be assessed as an exploratory question in its own right.

Ambiguity typing and persona mapping. The proposal anticipated manual inspection of a subset of texts, thematic clusters and persona pairs assigned consistently within each cluster. The realised classification instead used a fixed consensus threshold and a recorded pilot persona-flip decision for the remaining texts; persona pairs were assigned per text. The pilot judgments enter the category definition, so the author-relevant category is partly selected for the property tested by H3 (Section 3.4). A cluster-to-persona mapping was not produced under this per-text design. The realised pairings and rationales require the master sampling file to inspect; they are not tabulated in this manuscript.

Asymmetry in framing conditions. Models were evaluated under all four framings; humans were annotated under three, omitting the generic human condition. The control analysis in Section 5.10 is therefore available for models only, and no equivalent human decomposition of author-presence versus persona-content is possible. This asymmetry is a limitation of the collected data and is revisited in Section 6.3.

Inter-annotator agreement. The proposal specified Fleiss’ κ [Fleiss 1971]. Respondent identifiers are not required for this statistic: each item contributes its counts of emotion votes. Since the realised number of votes per (text, framing) cell ranges from three to seven, the conventional equal-rater calculation is restricted to the 754 cells with exactly three votes. The other 146 cells are excluded from this descriptive calculation only, not from the primary distributional analysis. Kappa is computed separately within each ambiguity pool and framing condition across the eleven emotion labels (Section 5.19). It adjusts for the group-specific marginal emotion frequencies, so comparing values between groups is descriptive rather than a causal test of framing. The equal-three restriction can also select a non-representative subset of persona cells.

Model evaluation metric. The proposal specified macro-F1 and percentage agreement against human majority labels. Section 5.4 supplies both as a secondary neutral-framing benchmark against the original corpus’s reader majority, restricted to the 241 sampled texts with a strict majority among five readers. Scores are calculated separately for each of the six prompt configurations and averaged by model; invalid labels are excluded and the number of retained predictions is reported in the reproducible table. The benchmark answers a narrower question than the primary analysis: it treats a noisy reader majority as the gold label, excludes 59 texts with no strict majority, and does not score persona responses against the separately collected human votes. Model framing behaviour is

instead characterised distributionally: by agreement with human vectors at baseline (Section 5.6), and by shift magnitude and direction under framing (Sections 5.9, 5.12).

Qualitative case analysis. The proposal specified an examination of individual strong-shift and stable-text cases. One traceable contrast, selected by its observed human shifts and reported in Section 5.14, illustrates how a single text can have a complete change under one persona and no change under another. This illustrative example is not a systematic thematic coding of cases and cannot establish why the shift occurred.

Explanations and corpus summaries. All seven analysed models provided a nonempty explanation for every one of their 7,200 responses (coverage is documented in `results/model_explanation_coverage_verified.csv`); an illustrative response is discussed in Section 5.14. No explanation-based thematic analysis or model confidence analysis was carried out. Section 5.2 gives author-label frequencies and text lengths for the final 300-text sample. These sample statistics do not stand in for a separately requested descriptive analysis of the full source corpus.

Participant count. The proposal anticipated 10–15 participants. The realised study drew on a larger, less formally bounded pool recruited across several collection rounds (an initial round and a later persona/neutral re-collection round, Section 4.1.5), with a target of three annotators per (text, framing) cell. Because respondent identifiers were not retained (Section 4.2), the exact number of distinct participants cannot be reconstructed after the fact, and no claim about it is made. This is itself a limitation, revisited in Section 6.3. The within-text permutation null in Section 4.6.2 does not require knowing how many people there were in total, but different people rating the two conditions removes only the within-text pairing requirement; it does not by itself establish that the conditions are exchangeable (Section 4.6.2).

4 Implementation

This chapter documents the full analytical pipeline, from the construction of prompts through to the statistical procedures used to test each hypothesis. It is written to be reproducible: every transformation applied to the data is described, together with the decisions taken where more than one treatment was defensible.

The pipeline has five stages. Prompts are constructed and models are queried (Section 4.1); human annotations are collected under the same materials (Section 4.2); every judgment, human and model alike, is converted into an eleven-dimensional probability vector (Section 4.3); metrics of shift are derived from those vectors (Section 4.4); and a quality-control gate (Section 4.5) must pass before any inferential statistic is computed (Section 4.6).

4.1 Prompt Construction and Model Evaluation

4.1.1 Stimulus Construction

Each stimulus consists of the unmodified text followed by an optional author cue, in the fixed format "Text: {text}" for the neutral condition and "Text: {text}\nAuthor: {cue}" for the other three, where {cue} is the literal string "a person" for the generic human condition and the assigned persona description for the two persona conditions. The text itself is byte-identical across all four framing conditions; only the cue, and its presence or absence, varies.

One property of the source corpus required correction during implementation. crowd-enVent texts were elicited with the frame "*I felt [emotion] when/that/if...*", and in a subset of texts the author's own emotion word remained present in the stimulus. Any such text leaks its answer to the reader. All affected texts were redacted so that the emotion word is replaced by an ellipsis, preserving sentence structure while removing

the answer. The detection and correction of this issue, which occurred after an initial round of model collection, is documented in Section 4.1.5.

4.1.2 Model Selection

Seven large language models were evaluated, spanning proprietary, commercial and open-weight systems: **OpenAI**, **Claude**, **Gemini**, **Qwen3**, **Gemma3**, **Ministral** and **Llama 3.1**. The proprietary and commercial models were queried through their respective APIs; the open-weight models were run locally through Ollama. The shared collection script also supported a Moonshot Kimi K3 endpoint as a fourth proprietary/-commercial option, but no Kimi K3 output was produced or retained: the corresponding `KimiK3_label/KimiK3_reason` columns are empty in all 7,200 rows of the raw result files, and a dedicated quality-control check confirms Kimi K3 has no normalised-label column and is correctly absent from the seven-model set analysed throughout this thesis.

Recovered model identifiers. The collection scripts’ configured defaults (not logged per-row in the result files themselves, so recovered from the scripts and their run logs rather than from the data) give: `gpt-5-nano` (OpenAI), `claude-haiku-4-5-20251001` (Claude), `gemini-3.6-flash` (Gemini), `qwen3:8b`, `gemma3:4b`, `ministral-3:8b` and `llama3.1:8b` (open-weight, via Ollama). No explicit temperature, top-*p*, or other decoding parameter was set for any model, so each used its provider’s or Ollama’s own default, which is not separately recorded and cannot be recovered. Collection dates are only partially reconstructable: run logs for the open-weight models place their collection in early September 2026 (specifically 2026-09-03 to 2026-09-04); no equivalent timestamped log was found for the proprietary and commercial models’ collection runs. These identifiers are the configured defaults actually present in the scripts that produced this data; they are not independently confirmed against a per-response API version field, since none was retained.

Using seven systems rather than the three required by the proposal serves a specific purpose for RQ2. A framing effect observed in a single model could reflect an idiosyncrasy of one training pipeline. An effect that replicates across seven independently developed systems is considerably harder to attribute to any one of them.

4.1.3 Prompt Configurations

Six prompt configurations were used, formed by crossing three prompt variants with two label formats:

- **Variants:** zero-shot, one-shot, few-shot.
- **Label formats:** the eleven emotions presented as a numbered list, or embedded inline in the question.

Each model answered every (text, framing) pair once under each configuration, giving $300 \times 4 \times 3 \times 2 = 7,200$ cells per model-ensemble position. Configurations are **never pooled at the level of raw responses**. Where an aggregate across configurations is reported, each configuration is computed independently first and the results are averaged afterwards, so that no analysis treats six correlated observations of one text as six independent observations.

4.1.4 Response Normalisation and Off-List Answers

Models were instructed to return exactly one emotion from the fixed set of eleven. Not all responses complied. Two classes of non-compliance were handled explicitly.

Formatting variation, responses differing only in case, whitespace or surrounding punctuation, was normalised to the canonical label.

Off-list answers, responses naming an emotion outside the eleven such as *anxiety*, *disappointment* or *frustration*, were **excluded** rather than mapped onto the nearest in-set label. Mapping would require adjudicating which of the eleven a given synonym corresponds to, and that adjudication would itself be a substantive interpretive judgment inserted into the data by the analyst. Exclusion is the conservative choice: it reduces the effective ensemble size for the affected cell rather than inventing a vote.

Off-list rates vary substantially by model and are reported in Section 5.3. No synonym-mapping sensitivity result is documented in the present analysis; whether the exclude-only policy affects a specific conclusion remains untested here.

4.1.5 A Data-Quality Correction Made During Analysis

Transparency requires recording corrections made partway through the study rather than presenting the corrected dataset as though it had always existed. The emotion-word leak described in Section 4.1.1 was caught and repaired in two separate rounds, at different scopes, and both are recorded here rather than only the more visible second one.

First round. An early audit of the stimulus texts identified 401 candidate texts still carrying the author’s emotion word, of which 102 fell inside the 300-text sample (78 corrected in place, 24 replaced outright because a clean substitute was not otherwise available). The model side was fully re-run against the corrected corpus. Because the human side had already been annotated for 78 of those 102 texts before the fix, those human ratings answered a stimulus that no longer existed; they were discarded and re-collected, together with the human annotations for the 24 replacement texts, in a dedicated collection round. The same round also collected the study’s first human ratings for the neutral condition, which up to that point had been supplied by reusing the original corpus’s five-reader labels, a different, disjoint rater pool from the persona conditions. All of this, 900 human vectors across three framings, each now from data collected specifically for the post-correction 300-text sample, was in place before the second round below began.

Second round. After an initial round of *model* collection against that corrected corpus, a further 49 of the 300 texts (16.3%) were found to still retain the author’s emotion word in the model stimulus, and the same texts additionally carried an outdated persona pair from an earlier design revision. The corrected wording existed in the master sampling file but had never been propagated into the model prompt files for that batch. This is a distinct defect from the first round rather than a recurrence of it: it affected the *model* prompt files specifically, was caught only after the human side described above was already complete, and is not claimed to share a root cause with the first round beyond both originating in the same elicitation frame.

The correction was made by **full re-collection**, not by exclusion. The stimulus and both personas were rebuilt from the corrected master file for all 1,176 affected rows (49 texts $\times$ 24 conditions), previous answers for those rows were discarded because they answered a prompt that no longer existed, and all seven models were re-queried.

This correction moved the results materially, which is precisely why it is reported. Model persona sensitivity rose from 8.7% to 11.3% on unambiguous texts and from 12.2% to 14.8% on author-independent texts, while author-relevant texts, of which only nine were affected, barely moved. The size of the change tracks the number of affected texts per category, which is consistent with the correction having removed a suppressing artefact, but this round changed two things for the affected 49 texts at once, both the leaked emotion word and the persona pair itself, so the rise cannot be attributed to removing the leak alone; some or all of it may instead reflect the updated persona pairs being more effective manipulations than the outdated ones they replaced. Every hypothesis verdict remained qualitatively unchanged.

4.2 Human Annotation Study

Human annotations were collected using the same texts, the same eleven-label set and the same author cues presented to the models, so that the two sides are responding to materially identical stimuli.

Each (text, framing) pair received multiple independent annotations, with a target of three annotators per cell; some cells received more. Crucially, and as specified in the approved proposal, **no annotator saw the same text more than once**. A participant who rated a text under one framing never saw that text under another.

This design property is load-bearing for the statistical treatment in Section 4.6. Because the conditions of a given text were rated by disjoint sets of people, there is no individual-level pairing across conditions to preserve. The human comparison is therefore **between-subjects at the level of framing condition**. Disjoint raters are a documented property of the collection procedure; exchangeability under a null of no framing effect additionally requires that the groups assigned to each condition would, absent a framing effect, have produced the same distribution of judgments, which disjoint recruitment alone does not guarantee if allocation or the participant pool differed systematically across rounds (Section 4.6.2).

The neutral condition was collected in a separate, later round from the persona conditions. Only vote values were retained in the analysis pipeline; respondent identifiers were not preserved. The consequences of this for inference are stated in Section 4.6.

A traceability limit in the neutral collection. The neutral condition’s forms were initially designed and distributed for one rater per text, and the collection records available alongside this thesis’s data reflect that design. The dataset used throughout this thesis instead carries three raters per neutral cell, matching the persona conditions. The origin of the additional two raters per text for the neutral condition could not be reconstructed from the retained collection records at the time of writing. This is disclosed here as a limit on this thesis’s own reproducibility rather than a claim that the votes are unreliable: the three-rater neutral vote counts are exactly what every downstream calculation in this thesis, including all six H1 tests, is computed from, and nothing in the analysis pipeline is inconsistent with them. But a reader attempting to trace every human vote back to a raw, timestamped response should know that this specific step in the collection history is not fully documented in the materials retained for this thesis.

4.3 Emotion Probability Vector Construction

This section describes the representational core of the thesis.

4.3.1 Definition

Let $E = \langle e_1, \dots, e_{11} \rangle$ be the fixed, alphabetically ordered emotion set. For any group of judgments about a single (text, condition), whether those judgments come from human raters or from models, let c_i be the number of judgments naming emotion e_i , and let $n = \sum_i c_i$ be the number of usable judgments. The emotion probability vector is

$$\mathbf{v} = \left\langle \frac{c_1}{n}, \frac{c_2}{n}, \dots, \frac{c_{11}}{n} \right\rangle, \quad \sum_{i=1}^{11} v_i = 1. \quad (4.3.1)$$

Every vector in this thesis is constructed this way. A human vector for a (text, framing) pair aggregates that cell’s raters; a model ensemble vector for a (text, framing, variant, format) cell aggregates the seven models’ single answers. Off-list and missing answers are excluded from c_i and reduce n accordingly; they are never silently counted as zeros across the board, which would distort the denominator.

4.3.2 Why Distributions Rather Than Labels

The case for this representation is empirical, not aesthetic. In the original five-reader corpus, only 80.3% of texts have a unique emotion commanding a strict majority. Of the remainder, 6.3% have a unique top choice that never reaches 50% (a plurality, not a majority) and 13.3% have **no unique top choice at all**. For roughly one text in five, a pipeline that reports a single label must either retain the most frequent label while concealing how divided readers actually were, or break a tie arbitrarily.

A vector representation avoids the choice entirely. It also makes a distinction available that a label cannot express: the difference between a judgment that barely moved and one that was substantially reinterpreted while happening to retain the same most-popular answer. Quantifying that difference is the subject of Section 5.6.

4.4 Metrics

Two families of metric are derived from the vectors: one for the *magnitude* of a shift and one for its *direction*. They answer different questions and are never conflated.

4.4.1 Magnitude: Total Variation Distance

For two vectors $\mathbf{p}$ and $\mathbf{q}$ over the same emotion set, total variation distance is

$$\text{TV}(\mathbf{p}, \mathbf{q}) = \frac{1}{2} \sum_{i=1}^{11} |p_i - q_i| \in [0, 1]. \quad (4.4.1)$$

TV is the share of probability mass that must be moved to turn one distribution into the other. It is zero when the two are identical and one when their supports are disjoint. Shift magnitudes are reported as $\text{TV} \times 100$ and referred to as shift percentages.

TV is a magnitude-only measure. It says how much moved, not which emotions moved or in which direction, which is why the directional metrics below are reported alongside it rather than as an optional addition.

4.4.2 Direction: Delta Vectors and Cosine Similarity

The signed change induced by framing is the delta vector

$$\delta = \mathbf{v}_{\text{persona}} - \mathbf{v}_{\text{neutral}}, \quad (4.4.2)$$

whose components sum to zero and record which emotions gained mass and which lost it. To ask whether a model's shift resembles the human shift *for the same text*, the two delta vectors are compared by cosine similarity:

$$\cos(\delta_H, \delta_L) = \frac{\delta_H \cdot \delta_L}{\|\delta_H\| \|\delta_L\|}. \quad (4.4.3)$$

Cosine is undefined when either delta is the zero vector, which occurs whenever one side did not move at all. These cases are **counted and reported** rather than silently dropped, because their frequency is itself a finding (Section 5.16).

4.4.3 Dominant Sets and Their Five Outcomes

The dominant set of a vector is the set of emotions attaining its maximum probability. It is a *set* rather than a single label because ties are common at small n and resolving them arbitrarily is exactly the practice this thesis is examining.

Comparing dominant sets before and after framing yields five descriptive outcomes:

- the sets are **identical**;
- a single emotion **split into a tie**;
- a tie **resolved** to a single emotion;
- two ties **partially overlap**;
- the sets are **disjoint**.

These describe relationships between complete top sets. A selected-label analysis instead chooses one winner from each set, so its changed/unchanged decision does not coincide with any one of these five outcomes; a tie can change the selected winner while the sets overlap. The set outcomes are reported in Section 5.17 and the selected-label comparison in Section 5.6.

4.4.4 Dispersion Measures

Three further quantities summarise the shape of a vector: Shannon entropy (and its normalised form), maximum probability as a measure of consensus strength, and support size, the count of emotions receiving any mass. These address a question distinct from magnitude, namely whether framing *concentrates* or *disperses* interpretation, and are analysed in Section 5.15.

4.5 Quality-Control Gate

No inferential statistic in this thesis was computed before its inputs passed quality control. Following the notebook rebuild described in Section 4.6.6, this is enforced at two levels rather than by a single external report. First, the data-loading routine that every notebook calls halts immediately, by assertion, if a structural invariant fails: there is no code path by which an analysis can run against malformed inputs. Second, an independent verification script re-derives 30 scientific and execution checks from the raw inputs and confirms that all three notebooks executed top to bottom with zero errors and consecutive execution counts; all 30 pass.

The checks fall into four groups:

- **Completeness:** every text present in every derived table; expected row counts exactly met (900 human vectors yielding 600 comparisons across two persona sides, 3,600 model comparisons across two persona sides and six prompt configurations, exactly six configuration rows per text and persona side).
- **Mathematical validity:** every probability vector sums to one; every delta vector sums to zero; total variation equals the sum of positive signed change; the retained vote count for every model comparison is equal on both sides of the comparison.
- **Design integrity:** exactly 100 texts per category; at least three model identities common to both sides of every model comparison; no duplicate (text, condition) rows; every one of the 300 sampled texts reproduces the classification rule of Section 3.3.1 exactly against the 985-text eligible candidate pool.
- **Source consistency:** every category value maps to a known category, and no text is classified inconsistently across its own rows.

The last group exists because of a defect it was written to catch. The human annotation file disagrees with itself on nine texts: their neutral rows record one category and their persona rows another, a residue of a reclassification between collection rounds. Because a naive de-duplication resolves such a conflict by whichever row happens to come first, those nine texts had silently entered the human-side analyses under the wrong category, producing an unbalanced 91/100/109 grouping in place of the designed 100/100/100. The fix is structural rather than a pinned exception list: the per-row category column in the human annotation file is no longer read at all, and every text’s category is instead looked up once, from the master sampling file, at data-loading time. The nine texts are verified here to confirm the defect is real and correctly resolved, not to certify a list that could silently go stale.

4.6 Statistical Procedures

This section states the inferential machinery and, equally importantly, the unit of analysis for each test.

4.6.1 Unit of Analysis

The recurring hazard in this dataset is pseudoreplication. A single text contributes 2 human comparisons and 12 model comparisons (2 persona sides $\times$ 6 configurations). Treating those as independent observations inflates n by up to twelvefold and correspondingly inflates significance.

Unless explicitly stated otherwise, **the unit of analysis is the text**. Where a statistic is computed over finer-grained rows, the confidence interval is obtained by a bootstrap that resamples *texts* rather than rows, so that the correlation among a text’s own observations is preserved in the resampling.

4.6.2 Permutation Tests

Two different permutation nulls are used [Good 2005], and the difference between them is substantive rather than technical.

For the human data (H1), votes are pooled within a text across the two conditions being compared, reshuffled, and re-split at the observed group sizes. Different people

rated the different conditions of a text, so no within-text participant pairing needs to be preserved. The permutation additionally assumes that, under no framing effect, the groups assigned to that text’s conditions would have the same distribution of emotion judgments; disjoint participants alone do not prove exchangeability if recruitment or allocation differed systematically.

For the model data (H2), the same pooling would be invalid. The same seven models answer both conditions, so model identity *is* tracked on both sides, and the models are not exchangeable with one another; Section 5.9.1 measures directly how differently they behave. The correct null holds model identity fixed and asks whether each model’s own two answers could equally well have been produced the other way round. The test is therefore a **paired sign-flip**: for each (cell, model) pair independently, the two labels are swapped with probability one half.

Both tests use 10,000 iterations with a fixed random seed.

4.6.3 Effect Sizes and Confidence Intervals

Effect sizes are reported alongside every p -value. Category contrasts (H3) are reported as the raw difference in mean shift between groups, with a text-level bootstrap confidence interval, rather than as a standardised ordinal effect size; matched-pairs rank-biserial correlation is used for paired comparisons (H4 magnitude), and Spearman’s ρ for monotonic association (H4 covariation). Confidence intervals are percentile bootstrap intervals with text-level clustering.

Where an association is summarised by a correlation, inference is drawn from the **clustered bootstrap itself** rather than from an analytic p -value computed over all rows. An analytic p over correlated rows is not merely imprecise, it is misleading: duplicating every row would leave ρ unchanged while shrinking p , which is exactly the behaviour clustering exists to prevent.

4.6.4 Multiple Comparisons

Two correction regimes are applied, and which applies is stated at each test. **Holm** correction [Holm 1979] is used for confirmatory hypothesis families, where control of the family-wise error rate is appropriate. **Benjamini–Hochberg** false discovery rate control [Benjamini and Hochberg 1995] is used for exploratory families such as the 66 per-

emotion tests (Section 5.13), where the goal is to limit the proportion of false positives among discoveries rather than to exclude them entirely.

4.6.5 Referencing Against a Sampling Null

A final procedure is required by a property of small-sample TV that is easy to overlook.

TV between two small vote samples is not zero when there is no effect.

Two independent three-vote draws from the same underlying distribution will frequently disagree, and they disagree *more* when that distribution is dispersed. Since the ambiguous categories are by construction the dispersed ones, a raw comparison of TV across categories measures baseline dispersion and any framing effect together.

To separate them, each text is given its own reference point. That text's votes are pooled across the two conditions, reshuffled, and re-split at the observed sizes; the average TV over many such draws is the *TV expected for that specific text* if framing did nothing. The text's **excess TV** is then the observed value minus this expectation.

This procedure is applied as a sensitivity analysis in Section 5.11.2. It is reported alongside the raw contrasts and does not replace them, because it was specified after the raw result was known and promoting it to primary status would constitute selecting an analysis on the basis of its outcome.

4.6.6 Reproducibility

All permutation tests and bootstrap procedures use a fixed seed and were verified to reproduce exactly across separate process invocations. One non-determinism defect was found and fixed during development: an unsorted set intersection produced an iteration order dependent on per-process hash randomisation, which made seeded results irreproducible between runs. Any collection of string keys feeding a seeded loop is now explicitly sorted before use.

The analysis notebooks were subsequently rewritten so that the calculations underlying the four main hypotheses are contained in the notebook itself rather than imported from an external module, to make that core pipeline auditable without cross-referencing separate source files. Every one of the 44 result tables produced by the rewritten notebooks was checked against the versions produced immediately beforehand and found byte-identical; the rewrite changed how the calculations are presented, not what they

compute. A small number of supplementary results added later (the inter-annotator agreement figures, the conventional majority-label benchmark, the entropy hypothesis tests of Section 5.15, and the transition-matrix figure of Section 5.17) are instead produced by standalone scripts alongside the notebooks rather than inlined into them.

5 Results

This chapter follows the question that motivated the analysis. First, it treats each text or condition as one selected emotion and checks what that method says about H1–H4. Next, it shows the shifts that this single label misses. Finally, it repeats the hypothesis tests using the full eleven-emotion distribution and explores which emotions gained or lost support. The statistical checks answer whether a visible pattern is distinguishable from variation expected when only a few people rate each condition; they do not replace the substantive question.

The original analysis numbers are computed by the executed analysis notebooks; the added agreement, corpus-description, explanation-coverage and text-level correlation checks have reproducible scripts and CSVs in the accompanying package. Effect sizes accompany every significance test, and results that failed to reach significance are reported as such rather than omitted.

5.1 Disagreement Is Present Before Any Manipulation

The premise of this thesis is that emotion judgments frequently admit no single correct answer [Pavlick and Kwiatkowski 2019; Uma et al. 2021]. It is worth establishing that this is true of the source corpus before any experimental manipulation is applied.

Each vector was classified into exactly one of four mutually exclusive vote patterns: *unanimous* (all judgments on one emotion), *majority* (one emotion holds a unique, strict majority above 50%), *plurality without majority* (one emotion has the unique top share but at or below 50%), and *tied* (two or more emotions share the top share).

Interpretation. Among the 300 sampled texts, 80.3% have a unique emotion with a strict majority (71 unanimous plus 170 non-unanimous majority texts, i.e. 241 of 300). This sample is stratified two-thirds toward ambiguous categories by design (Section 3.3.2), so this rate does not characterise the source crowd-enVent corpus at large,

Table 5.1: Vote-pattern distribution across the three judgment sources.

Pattern	Original corpus (up to 5 retained votes, 300 texts)	New human (900 vectors)	Model ensemble (7 models, 7,200 vectors)
Unanimous	23.7%	18.3%	36.8%
Majority (non-unanimous)	56.7%	49.0%	53.3%
Plurality, no majority	6.3%	4.8%	5.8%
Tied	13.3%	27.9%	4.1%

only the sampled texts analysed here. The remaining **19.7%** do not have a strict majority: of these, 6.3% still have a unique plurality label, so a pipeline reporting a single label would apply a plurality rule that overstates agreement, while the other 13.3% have no unique top choice at all and would require breaking a tie arbitrarily.

Two further observations follow. The new three-rater human data ties far more often (27.9%), which is an arithmetic consequence of small n rather than evidence that the new annotators disagreed more. The model ensemble ties least (4.1%) and is unanimous most often (36.8%), indicating that the seven models converge more readily than human readers do, a difference that recurs throughout this chapter.

5.2 Sample Composition

In the final sample of 300 texts, the original author’s emotion labels occur as follows: anger 32, disgust 35, fear 27, guilt 14, joy 25, pride 21, relief 25, sadness 29, shame 19, surprise 33 and trust 40 (`results/sample_author_emotion_frequencies_verified.csv`). The text length has a median of 13 whitespace-separated words (interquartile range 8–25; minimum 2, maximum 203). These are *sample* summaries after category balancing and eligibility screening, not frequencies for the original full crowd-enVent corpus.

5.3 Model Response Validity

Of 50,400 individual model judgments ($7 \text{ models} \times 7,200 \text{ cells}$), **609 (1.21%)** were excluded as off-list or missing. Exclusion rates vary by an order of magnitude across models. These counts are computed directly from `experiment_results_all.csv` using the same label-resolution logic the analysis pipeline applies (normalisation, then exclusion of any response outside the eleven-emotion set); they summarise an input property

of the data rather than an analysis result, which is why they are reported here rather than derived in a results notebook.

Table 5.2: Excluded (off-list or missing) answers by model, out of 7,200 judgments each.

Model	Excluded	Rate
OpenAI	0	0.00%
Gemini	0	0.00%
Ministral	1	0.01%
Qwen3	8	0.11%
Claude	80	1.11%
Gemma3	93	1.29%
Llama 3.1	427	5.93%

Interpretation. Llama 3.1 accounts for 70% of all exclusions, most often by answering with a near-synonym outside the eleven-label set: *anxiety*, *disappointment*, *frustration*, *embarrassment*. This is a genuine behavioural difference in instruction-following, not a data defect. Because off-list answers reduce the effective ensemble size for the affected cell rather than being mapped onto a nearest label, no interpretive judgment is inserted by the analyst; the cost is a slightly smaller n for those cells.

5.4 Conventional Majority-Label Benchmark

To provide the proposed classification metrics alongside the distributional analysis, the neutral response from each model is compared with the original corpus’s stored reader-majority label. A strict majority (at least three of five readers) exists for 241 of the 300 sampled texts; the other 59 are excluded so that a tie or non-majority plurality is not silently treated as a gold standard. Percentage agreement and eleven-class macro-F1 are calculated on the valid model answers within each prompt configuration and then averaged equally over the six configurations. Thus each number in Table 5.3 is descriptive rather than a test comparing models; per-configuration scores and retained counts (217–241 of 241) appear in `results/neutral_majority_label_benchmark_verified.csv`.

These numbers show agreement with one deliberately narrowed, relatively consensual neutral reference set. They do not measure similarity of full emotion distributions, generalise to the 59 excluded texts, or test whether a model’s framing response matches the human shift.

Table 5.3: Neutral model performance against the original readers’ strict majority, averaged over six prompt configurations.

Model	Agreement	Macro-F1
OpenAI	77.0%	0.742
Claude	73.8%	0.715
Gemini	78.8%	0.764
Qwen3	70.2%	0.686
Gemma3	63.2%	0.585
Ministral	62.3%	0.611
Llama 3.1	68.8%	0.650

5.5 First Pass: H1–H4 with One Selected Emotion

The old approach chooses one top emotion for each condition (using a deterministic tie-break), then records whether that selected emotion changes when a persona is supplied. This answers a plain question: *did the reported label change?* It cannot measure smaller redistributions among the eleven possible emotions.

Table 5.4: First-pass conclusions using selected emotion labels only. The full calculations are in the accompanying old-method CSVs.

Question	Selected-label result
H1, people	None of the six category-by-persona tests is significant after Holm correction. Some labels change, but the evidence is insufficient to detect a human framing effect by this method (<code>old_H1_H2.csv</code>).
H2, models	Both overall persona contrasts are significant ($p_{\text{Holm}} = 0.0002$ each). A model-ensemble effect is detectable even with one label (<code>old_H1_H2.csv</code>).
H3, category	The directly predicted author-relevant minus author-independent difference is 4.50 percentage points and not significant ($p_{\text{Holm}} = 0.472$; <code>old_H3_raw.csv</code>).
H4, alignment	Human label-change frequency exceeds the model-ensemble frequency by 34.44 percentage points; 102 of 438 pairs where both change have the same exact selected-label transition (<code>old_H4_frequency.csv</code> ; <code>old_H4_change_agreement.csv</code>). Direction can be described only as agreement between discrete transitions.

The first-pass result is useful, including its negative findings. The question is whether it mistakes a stable winner for a stable interpretation. The next section measures that loss directly, using the same texts and responses.

5.6 What the Vector Representation Preserves

This section quantifies the claim motivating the entire representational choice. Every comparison was classified by what a dominant-label pipeline would have recorded versus what the distribution actually did.

Table 5.5: What a selected-label analysis would record. These three outcomes are exhaustive and mutually exclusive; only the middle row is invisible to a label-based pipeline, which records it as unchanged despite the distribution having moved.

Outcome	Human	Model ensemble
No change at all (TV = 0)	8.5% (51)	40.2% (1,446)
Moved with the same selected label (proportion-only or support change)	<i>invisible to label analysis</i> 38.7% (232)	41.4% (1,492)
Winner changed (recorded as changed)	52.8% (317)	18.4% (662)

Interpretation. A selected-label pipeline would record 52.8% of human comparisons and 18.4% of model comparisons as changed (`results/old_H4_frequency.csv`). It would additionally file **38.7% of human** and **41.4% of model** comparisons (232 and 1,492 comparisons respectively) as *unchanged*, despite the distribution demonstrably having moved.

For the model ensemble in particular, the comparisons invisible to a label-based analysis outnumber those it detects by more than two to one (1,492 versus 662). This is the concrete content of the claim that aggregation discards information, measured across all 4,200 valid comparisons rather than argued from examples. A stricter, tie-aware version of the same question, asking whether the *complete* set of tied top emotions changed rather than merely the alphabetically selected one, gives smaller “concealed movement” figures: 25.8% human and 38.7% model (`results/information_lost_by_labels.csv`), because it grants the label method credit for tracking ties. Section 5.17 reports a related but distinct breakdown, the full set-relationship outcomes between the two sides’ top-emotion sets (identical, disjoint, tie formed or resolved), rather than this same concealed-movement percentage.

A related result concerns human–model agreement at baseline. Under neutral framing, human and model dominant labels agree in 58.0% of comparisons overall, ranging from 74.2% on unambiguous texts down to 49.8%–50.0% on the two ambiguous categories. Within the matching and non-matching buckets, full-vector cosine similarity spans a wide range: comparisons whose labels agree have mean cosine 0.90 across [0.41, 1.00], and those whose labels disagree have mean cosine 0.41 across [0.00, 0.98]. A binary match

or mismatch therefore collapses a continuous and broad range of actual distributional similarity into one of two buckets, and even within the “agree” bucket the underlying distributions are far from identical.

Second pass: the same hypotheses with full emotion distributions. In the next four sections, a shift means that the mix of emotions changed, even if its most common label stayed the same. A distance of 20% means that one fifth of the total emotion probability would have to be moved to make the two distributions match. Because each human distribution contains only a few votes, some movement occurs by chance even when framing has no effect. The statistical tests compare observed movement with that baseline; the p -values do not measure how large or important an effect is.

5.7 Two Independent Neutral Measurements Disagree Substantially

Before any framing effect can be interpreted, it is necessary to establish how much two measurements of the *same* condition differ. The original corpus’s five-reader neutral round and this study’s own three-rater neutral round are two independent measurements of the same texts under no manipulation.

Table 5.6: Mean TV distance between the original corpus neutral round and this study’s neutral round.

Category	Mean TV
Unambiguous	33.1%
Author-Independent Ambiguous	47.5%
Author-Relevant Ambiguous	53.2%

Interpretation. Two “no persona, no manipulation” measurements of the same texts differ by 33.1% to 53.2% of probability mass, and they differ *more* in the categories designed to be more ambiguous. This is the noise floor against which all subsequent shift magnitudes must be read.

The implication is methodological and is applied throughout: a raw shift percentage is not interpretable on its own. A 45% human shift on unambiguous texts sits barely above a 33.1% baseline disagreement, and it is the permutation control, not the raw magnitude, that determines whether a framing effect is present. This consideration returns in Section 5.11.2 in a sharper form.

A second, non-overlapping baseline comparison is available for the neutral round specifically: human and model-ensemble dominant labels under neutral framing agree in 74.2% of unambiguous comparisons but only 49.8% and 50.0% of author-independent and author-relevant comparisons respectively, and the corresponding mean TV distances (32.1%, 48.0% and 49.6%) track the same ordering. Humans and models therefore start from noticeably different readings of the ambiguous categories even before any persona is introduced.

5.8 H1: Author Framing Shifts Human Judgments

Hypothesis. Human emotion judgments will differ significantly across author-framing conditions for ambiguous texts, but not (or substantially less) for unambiguous control texts.

5.8.1 Magnitude by Category

Mean text-level shift magnitude rises across the three categories as predicted: Unambiguous **45.1%** (median 41.7%), Author-Independent **54.1%** (51.7%), Author-Relevant **60.3%** (64.6%).

Figure 5.1 shows that the ordering is accompanied by heavy distributional overlap. Interquartile ranges are wide and individual texts span the full range in every category. The figure must not be read as showing that every author-relevant text shifts more than every unambiguous one; it shows a difference in central tendency against considerable within-category variation.

5.8.2 Permutation Test

Raw magnitudes are not interpretable against the noise floor established in Section 5.7, so each category and persona side was tested against a within-text permutation null (10,000 iterations, Holm-corrected across the six tests).

Interpretation. Three results, stated in the order of their evidential weight.

Author-relevant ambiguous texts show a significant framing effect on **both** persona sides. This is the core prediction of H1 and it is supported.

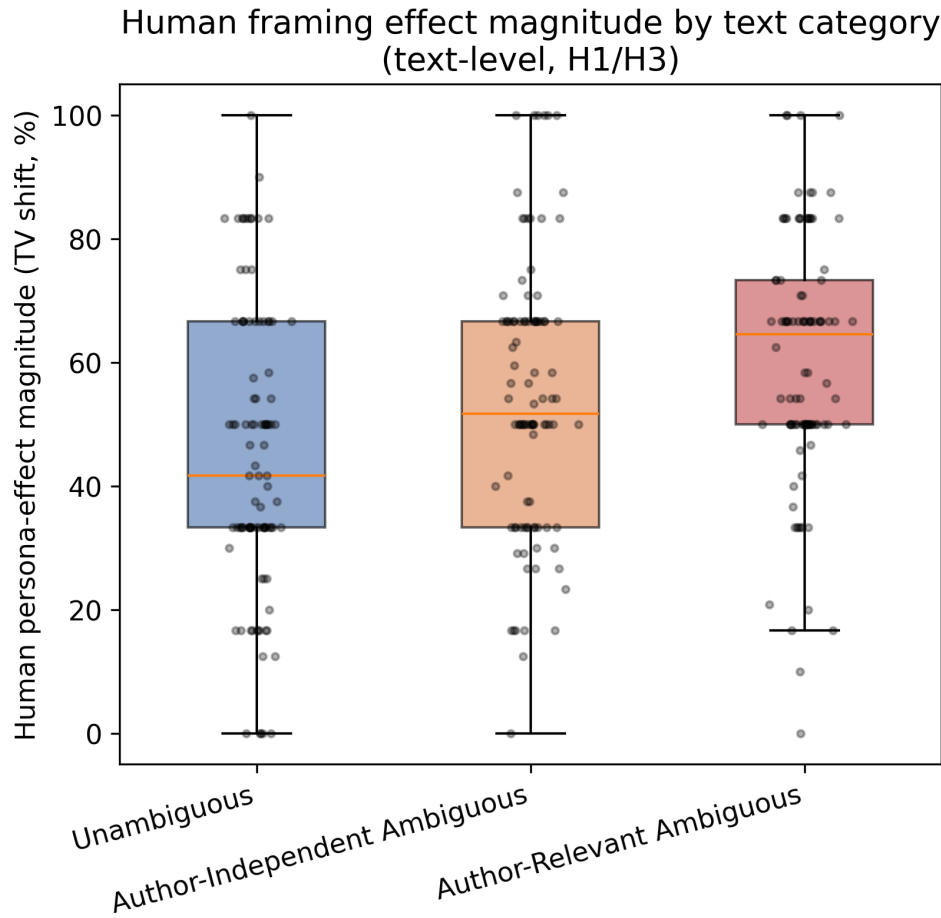


Figure 5.1: Human persona-effect magnitude by text category (text-level, one observation per text).

Unambiguous control texts show **no statistically detectable** effect on either side. This is consistent with H1’s control prediction, but non-rejection does not establish that the true effect is zero or that the manipulation operated exclusively through author content.

For author-independent ambiguous texts, side a crosses the significance threshold and side b does not. The observed shifts are 56.3% and 51.9%, against respective conditional-null means of 50.9% and 48.9%. *Significant on one side and non-significant on the other is not a test of a difference between sides.* An exploratory direct comparison pairs each text’s two raw TV shifts: the mean A-minus-B difference is 4.44 percentage points, with a text-bootstrap 95% interval $[-2.08, 11.06]$ points (paired sign-flip $p = 0.186$; see `results/author_independent_persona_side_contrast_verified.csv`). Thus the ev-

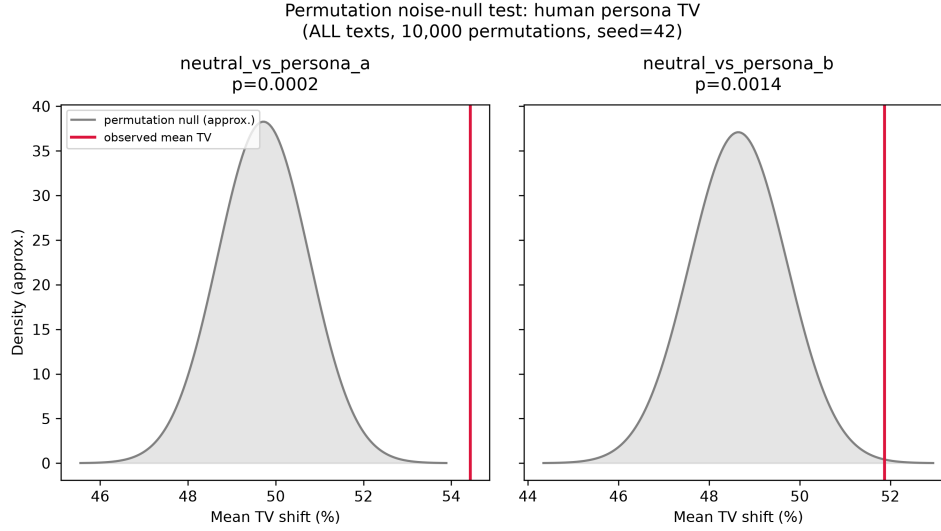


Figure 5.2: Permutation null distributions for the human persona effect, pooled across all texts, with the observed mean marked.

Table 5.7: H1 permutation test results. Holm-corrected across six tests (3 categories $\times$ 2 persona sides).

Category	Persona side a		Persona side b	
	Observed TV	p_{Holm}	Observed TV	p_{Holm}
Author-Relevant	59.9%	0.0066	60.7%	0.0112
Author-Independent	56.3%	0.0110	51.9%	0.1980
Unambiguous	47.1%	0.1980	43.0%	0.2145

idence is mixed about whether *each* framing effect is detectable; it does not establish that one persona has a stronger effect than the other. This exploratory comparison is of raw TV, not of the two separately estimated excess-over-null values.

A cross-check makes the case for the distributional representation directly rather than by argument. Repeating these same six comparisons on the *selected label* rather than the vote vector (pooling each text’s votes, permuting, and asking only whether the winning emotion differs, with the identical Holm correction) rejects the null in **0 of 6** cases under this thesis’s tie-breaking convention (a vote-share tie goes to whichever emotion is alphabetically first). Under the conventional one-label treatment of this same human data, no framing effect is detectable at all, in any category, on either persona side, under that convention. Nonzero label-change rates exist regardless (Section 5.6), but they do not pass the corresponding significance tests once each distribution is reduced to a single winner.

This exact count is itself sensitive to the tie-break rule. Ties are common at these sample sizes and the choice of which emotion wins a tie is arbitrary; there is no principled reason to prefer alphabetical order. Re-running the identical selected-label test with ties broken in reverse-alphabetical order instead changes the author-relevant, persona-A cell from non-significant to significant (change rate 73.0%, $p_{\text{Holm}} = 0.0012$), making it **1 of 6** rather than 0 of 6 (`results/tie_break_h1_significance_verified.csv`). The broader methodological point, that the selected-label method detects at most a fraction of what the distributional treatment detects, holds under either convention; the specific claim that it detects *none* of the six does not.

Verdict: H1 is partially supported: supported on author-relevant texts, with no detected effect in the unambiguous controls and mixed evidence on author-independent texts. That this support is visible only under the distributional treatment, and vanishes under the selected-label one, is itself among this chapter’s findings and is returned to in Section 5.6.

5.9 H2: Author Framing Shifts Model Classifications

Hypothesis. LLM emotion classifications will shift across author-framing conditions, indicating sensitivity to minimal author cues.

5.9.1 Magnitude by Category and Configuration

Mean ensemble shift is **11.3%** for unambiguous, **14.8%** for author-independent and **28.8%** for author-relevant texts. The ordering holds in every one of the six prompt configurations (Figure 5.4); each of the seven models also has a positive observed label-change rate. The paired test below applies to the model ensemble rather than testing each model separately.

Individual model label-change rates on author-relevant texts range from Ministral at 39.3% down to Gemma3 at 26.2%. No pairwise significance test was run between models, so these are described as highest and lowest *observed* means, not as best and worst.

Note the strong right skew visible in Figure 5.3: median ensemble shift is well below the mean, because a large share of model comparisons do not move at all (Section 5.16).

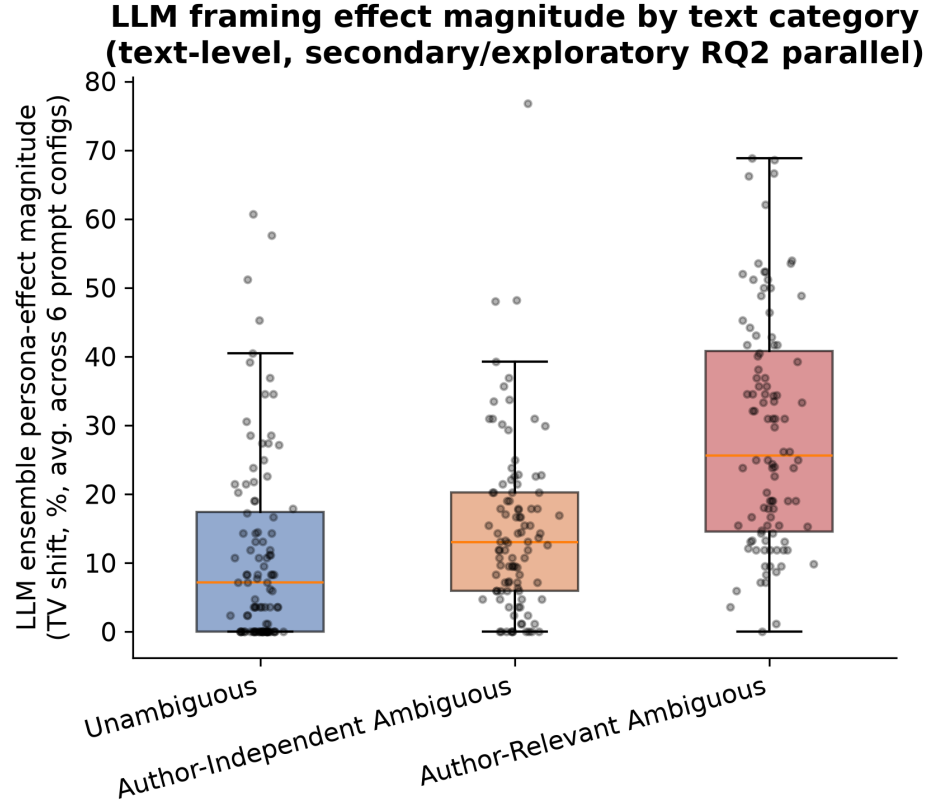


Figure 5.3: Model ensemble persona-effect magnitude by text category (text-level, averaged across the six prompt configurations).

5.9.2 Paired Permutation Test

Because the same seven models answer both conditions, the null must hold model identity fixed (Section 4.6.2). The paired sign-flip test gives:

Table 5.8: H2 paired sign-flip permutation test, Holm-corrected across two persona sides.

Comparison	Observed mean TV	95% CI (text-clustered)	p_{Holm}
Neutral vs persona a	16.80%	[14.74, 18.97]	0.0002
Neutral vs persona b	19.78%	[17.50, 22.08]	0.0002

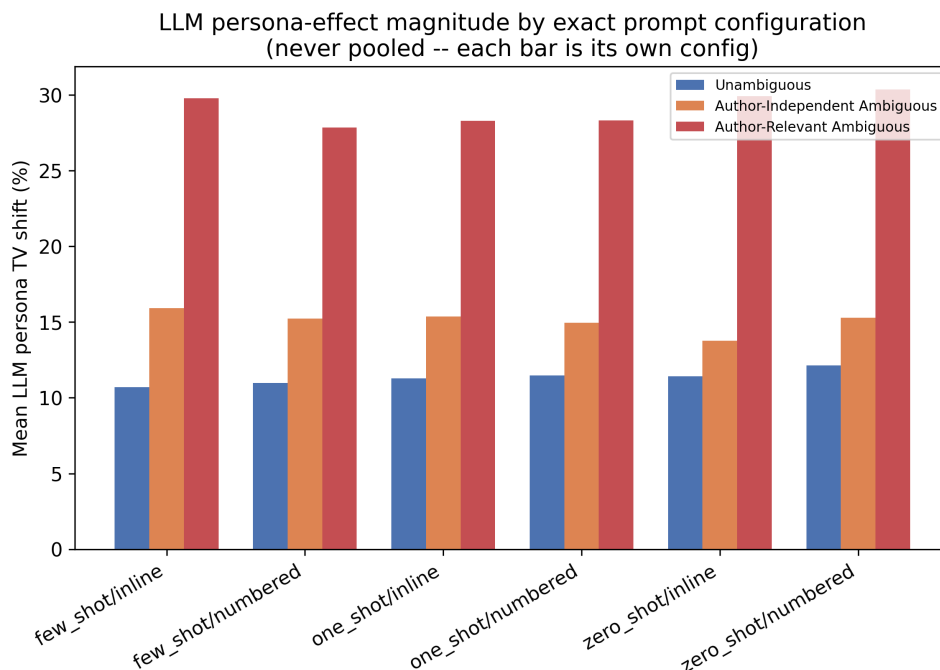


Figure 5.4: Model persona effect by exact prompt configuration. Each bar is a separately computed, never-pooled comparison.

5.9.3 What This Test Does and Does Not Establish

The statistic is TV between ensemble vectors, so the test detects **coordinated net redistribution** across models. It is not a test of whether any individual model changed its answer, and the distinction is not merely pedantic.

Under a paired sign flip, concordant model pairs cancel and only discordant ones contribute. If a cell contains exactly one discordant model, its TV equals $1/n$ under *every* possible swap: the null is a point mass, and the test cannot reject regardless of how real the change is. In these data, 37.9% of cells have no discordant model and a further 23.3% have exactly one, so **61.2% of cells contribute no variance to the null**. The test is driven by the remaining 38.8%.

The complementary quantity is therefore reported alongside it: of 24,736 valid model pairs, **5,263 (21.3%) changed their label** between conditions. H2 as worded is supported by both measures, but they answer different questions and only the first is what the p -value refers to.

Unlike H1, H2’s verdict does not depend on the choice of representation: the same selected-label cross-check described in Section 5.8.2 also rejects the null on both persona sides under the old, one-label treatment ($p_{\text{Holm}} = 0.0002$ for both). The model’s selected-label test survives aggregation to a single winner; most of the human selected-label tests do not, but not because the human raw change rate is smaller: at the label level, humans change their selected answer far more often than the model ensemble does (52.8% of human comparisons versus 18.4% of model comparisons, Table 5.5). The human tests are underpowered relative to the model tests, most plausibly because each human cell has only three votes against the model ensemble’s larger effective sample, not because the underlying human effect is weaker.

Verdict: H2 is supported on both persona sides.

5.10 Is It the Persona, or Merely Having an Author?

Every neutral-versus-persona comparison carries a confound: persona prompts add an author line that neutral prompts lack entirely. A measured shift could reflect the persona’s content or merely the presence of an attribution. The `generic_human` condition separates them.

Table 5.9: Decomposition of the model framing effect using the generic-human control. Each row uses only the models that answered validly under all three of the neutral, generic-human and persona conditions for that side (a three-way intersection, stricter than matching models pairwise), so the two component distances and the confounded distance are computed over the same model set.

Comparison	Side a	Side b	95% CI (a / b)
Neutral → generic human	7.98%	8.01%	[7.23, 8.75] / [7.26, 8.77]
Generic human → specific persona	16.20%	19.30%	[14.14, 18.33] / [17.01, 21.61]
Neutral → persona (<i>confounded</i>)	16.73%	19.78%	[14.68, 18.89] / [17.50, 22.08]

Interpretation. Supplying a generic author line alone produces a shift of essentially the same size on both sides (7.98%–8.01%), which is not negligible: naming any author at all changes how the models classify. But the effect of the persona’s specific content, measured with the author line held constant, is roughly twice as large on both sides and is essentially as large as the full confounded comparison. These three distances describe what is associated with each step under matched models; they are not claimed to be additive components of one total effect.

The persona-content step is the larger of the two measured distances on both sides, while the author-line-alone step is smaller but still non-negligible. This analysis is labelled secondary because the decomposition was not specified in the approved proposal, but it converts a limitation that could otherwise only be acknowledged into one that is measured.

5.11 H3: Framing Effects and Ambiguity Type

Hypothesis. Framing effects will be stronger for author-relevant ambiguous texts than for author-independent ambiguous texts.

5.11.1 Raw Contrasts

Each text is reduced to one number, its mean shift over the two persona sides, and the three categories are compared pairwise. The permutation null shuffles which category each text's number belongs to; the effect size is the raw difference in mean shift, and its confidence interval is a text-level bootstrap.

Table 5.10: H3 pairwise contrasts on raw TV, Holm-corrected across the three contrasts.

Contrast	Difference	95% CI	p_{Holm}
Author-Independent vs Unambiguous	9.04pp	[2.63, 15.33]	0.0144
Author-Relevant vs Author-Independent	6.18pp	[0.30, 12.08]	0.0403
Author-Relevant vs Unambiguous	15.22pp	[9.08, 21.29]	0.0003

All three contrasts are significant in raw terms after correction. The contrast that H3 is literally about, author-relevant versus author-independent, is the **weakest of the three**, with the smallest difference, the widest relative confidence interval, and the largest corrected p -value. It should not be presented as a dramatic separation, and, as Section 5.11.2 shows next, this specific contrast does not survive being referenced against each text's own sampling null, so "significant in raw terms" is the most this result supports on its own.

5.11.2 Referencing Against Each Text's Own Sampling Null

A property of small-sample TV complicates the raw comparison. As set out in Section 4.6.5, TV between two three-vote samples is not zero under no effect, and it grows

with the dispersion of the underlying distribution. The ambiguous categories are by construction the dispersed ones. A raw category comparison therefore measures baseline dispersion and framing effect together.

Table 5.11: Observed, expected-under-null, and excess TV by category (human, text-level).

Category	Observed TV	Expected under sampling null	Excess
Unambiguous	45.07%	43.07%	2.00pp
Author-Independent	54.10%	49.90%	4.20pp
Author-Relevant	60.28%	54.59%	5.70pp

The expected null TV is *itself* strongly ordered by category, from roughly 43% to 55%. Once each text is referenced against its own null, the excess above that conditional-null expectation is on the order of two to six percentage points rather than forty-five to sixty.

Table 5.12: H3 contrasts on raw TV versus null-referenced excess TV.

Contrast	Raw	Null-referenced
Author-Relevant – Author-Independent	6.18pp, $p_{\text{Holm}} = 0.040$	1.49pp, $p_{\text{Holm}} = 0.697$
Author-Independent – Unambiguous	9.04pp, $p_{\text{Holm}} = 0.014$	2.21pp, $p_{\text{Holm}} = 0.697$
Author-Relevant – Unambiguous	15.22pp, $p_{\text{Holm}} = 0.0003$	3.70pp, $p_{\text{Holm}} = 0.287$
Holm-significant	3 of 3	0 of 3

Interpretation, stated carefully. The ordering is preserved in direction under null-referencing ($2.00 < 4.20 < 5.70$), but none of the three contrasts remains significant after correction.

Three things follow, and the distinctions between them matter.

First, the raw contrasts show an observed difference motivated by the approved proposal. H3’s predicted ordering *is* present in observed shift magnitude.

Second, non-significance under null-referencing is **not** evidence that the categories are equivalent. The excess-TV estimator subtracts an estimated conditional-null quantity, and the resulting point estimates are far smaller (1.49–3.70pp versus 6.18–15.22pp raw) while the confidence intervals are, if anything, narrower in absolute width than the raw contrasts’ intervals (about 8.6–9.5pp wide versus 11.8–12.7pp raw). None of the three reaches significance because the estimated excess itself is small relative to that width, not because the interval has widened. Absence of evidence is not evidence of absence.

Third, excess TV measures observed distance above the chosen conditional-null expectation; it is a useful sensitivity statistic, not a validated decomposition into causal framing

and baseline-noise components, since the pooled votes it references contain observations from both conditions and their dispersion can itself reflect condition differences. What it does establish is that the category contrast becomes small and statistically inconclusive under this reference, so the raw ordering is sensitive to the choice of analytical reference, and this study cannot cleanly separate framing from baseline dispersion at 100 texts per category. A strong behavioural reading of the raw ordering is therefore not supported by these data.

This analysis was specified after the raw result was known. It is consequently reported as a sensitivity analysis alongside the raw contrasts and does not replace them; promoting it to primary status would amount to selecting an analysis on the basis of its outcome.

Verdict: H3 shows the predicted raw ordering, but the framing-specific contrast is inconclusive after referencing baseline vote dispersion. It should be stated as an ordering in observed disagreement, not as a demonstrated framing-sensitivity gradient. Recall also from Section 3.4 that the categories were partly defined using an LLM judge’s own sensitivity to author-relevance, which makes H3 a transfer comparison rather than an independent discovery of that grouping principle.

5.12 H4: Human–Model Alignment

Hypothesis. The direction of framing-induced shifts in LLM classifications will partially align with human judgment shifts, but the magnitude of these shifts will differ.

The two components are tested separately, because they make different predictions and one can hold without the other.

5.12.1 Magnitude

Two distinct questions must be separated here, and conflating them is a common error.

Covariation asks whether texts that shift more for humans also shift more for models. Spearman’s $\rho = 0.331$, 95% CI [0.223, 0.428], is computed on the 300 text-level means, with both persona sides averaged on the human side and both sides and six configurations averaged on the model side (text bootstrap; Figure 5.5). The calculation is recorded in `results/h4_text_level_covariation_verified.csv`. This positive association says nothing about whether the two magnitudes are equal.

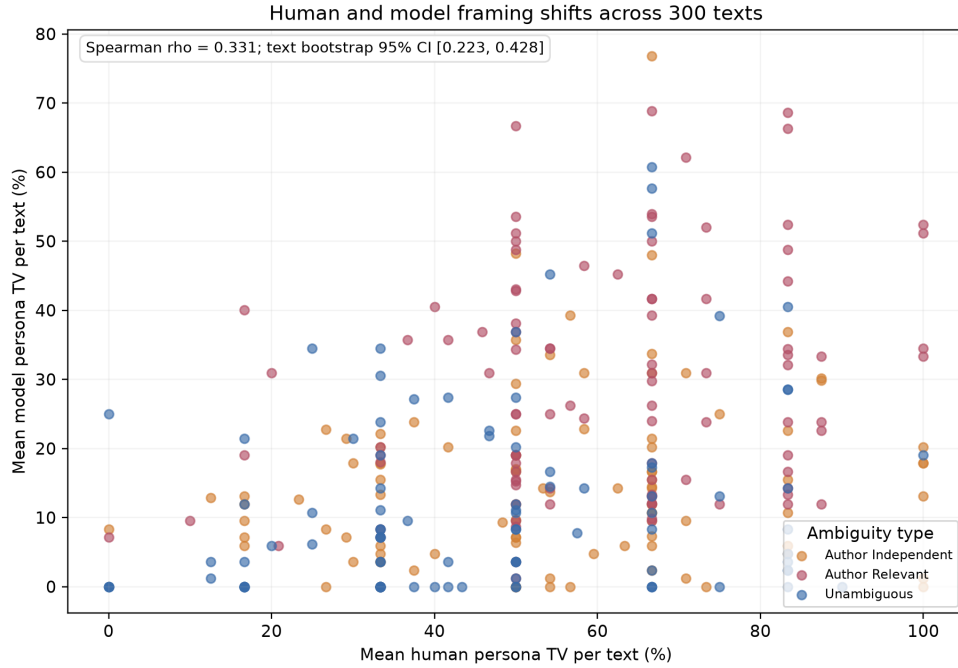


Figure 5.5: Mean human versus mean model shift magnitude for each of 300 texts, coloured by category.

Difference asks whether the magnitudes differ, which is what H4 actually predicts. A paired Wilcoxon signed-rank test over 300 text-level pairs gives a mean paired difference of **34.86pp** (human minus model), 95% CI [32.23, 37.54], $p = 4.68 \times 10^{-48}$, with matched-pairs rank-biserial $r = 0.978$, close to the theoretical maximum, meaning human shift exceeds model shift for nearly every text (279 of 300 texts move in that direction, 16 the other way, 5 tied), not merely on average.

5.12.2 Is the Magnitude Gap an Artefact of Vote Counts?

Human vectors are built from three raters and model vectors from up to seven models. TV is partly a function of how many votes underlie it, so some of the gap could be arithmetic rather than behavioural. This was tested directly rather than argued about.

Both sides were standardised to three votes: human votes were subsampled to three per condition, and the *same three model identities* were drawn for the neutral and persona sides of each cell, holding ensemble composition fixed as well as size. Two hundred

independent draws per comparison were taken, with the point estimate an equal-weight average across the balanced category $\times$ side $\times$ configuration strata.

Table 5.13: Magnitude gap before and after standardising both sides to three votes.

	Mean human – model gap
Unmatched	34.86pp
Count-standardised (3 votes vs the same 3 models)	$\approx$ 33pp
Reduction from count standardisation	$\approx$ 1.5pp (about 4% of the raw gap)

A note on precision. The standardised figure is a Monte Carlo estimate over 200 draws per comparison, and is quoted deliberately as *approximately* 33 percentage points rather than to two decimals.

The estimate is exactly reproducible: holding the inputs and the seed fixed returns the same value to six decimal places on repeated runs. What moves it slightly is the *ordering* of votes within a cell. Subsampling draws indices, so when a condition holds more than three votes, the same multiset presented in a different order yields different sampled subsets under a fixed seed. Re-ordering the votes within each cell, sorting or reversing them while leaving the multiset untouched, moves the estimate between 33.34 and 33.37pp, and five independent seeds span 33.32 to 33.37pp.

That spread, roughly 0.04pp, straddles the boundary at which a one-decimal figure would round either way, which is why no decimal is claimed. Nothing in the interpretation turns on it: count standardisation reduces the estimated gap by approximately 1.5 percentage points under every ordering and every seed tested.

Interpretation. Standardising both sides to the same vote count reduces the estimated gap from 34.86 to approximately 33 percentage points, a reduction of roughly 1.5 points (about 4%), and the difference survives within every category. Consistent with the note on precision above, this is stated to the nearest whole point rather than to two decimals, since the second decimal place moves by more than that between orderings and seeds. This procedure does not uniquely identify the share of the gap causally explained by vote counts, nor does it remove the between-person/within-model design asymmetry discussed below; humans still shift substantially more than the model ensemble, and this is not an artefact of counting.

Two qualifications remain and are not addressed by count standardisation. The human comparison is **between-person** (different raters in each condition) while the model comparison is **within-model**, since the same model answers both. Part of the hu-

man shift is therefore between-rater variability with no model counterpart. And each model produced only one generation per condition, so model-side stochastic variability is unmeasured.

5.12.3 Direction

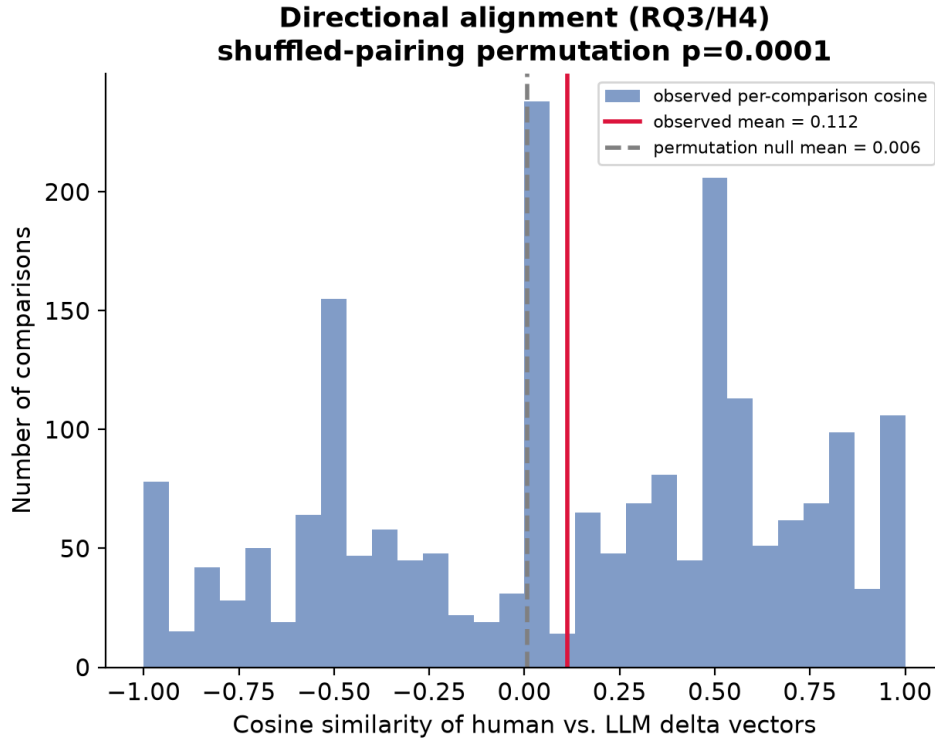


Figure 5.6: Distribution of per-comparison directional cosine between human and model delta vectors, one point per (text, side, prompt-configuration) comparison, with the observed mean and the chance-pairing null marked.

Mean cosine between human and model delta vectors is **0.112**, 95% CI [0.072, 0.152], against a shuffled-pairing permutation null (mean 0.006), $p = 0.0001$. The model ensemble’s shift direction is therefore related to the *specific* text’s human shift direction more than chance pairing would produce, but a cosine of approximately 0.11 is a small, partial association, not a close match.

Coverage. Persona sides are kept separate and never averaged together, so this is computed at the level of individual (text, side, prompt-configuration) comparisons through-

out, not on a coarser per-text average. Of the 3,600 such comparisons, **2,020 (56%)** have a defined cosine; 252 of the 300 texts (84%) have at least one defined comparison among their twelve. The undefined cases are those where one side’s delta is exactly zero, so there is no direction to compare, and they are excluded rather than treated as zero.

Verdict: H4 is partially supported. The magnitude component is **supported**: the magnitudes differ, substantially and robustly to count standardisation, which is what H4 predicts. The direction component is **partially supported**: alignment is above chance but weak, which is the literal content of H4’s word “partially”.

It should be noted that the direction analysis is not immune to small-sample noise. Cosine is invariant to multiplying a vector by a positive scalar, but drawing three votes rather than seven is not scalar multiplication: it can change a delta’s direction, and it can make the delta exactly zero, in which case direction becomes undefined rather than being preserved.

5.13 Where the Probability Mass Moved

The largest descriptive movements are a gain of **+4.0pp** for *guilt* in the author-independent category and a loss of **−5.4pp** for *sadness* in the unambiguous category, with *fear* gaining and *anger* losing elsewhere.

Extending the test to the model ensemble changes the outcome. The original analysis tested only the 33 (category × emotion) human contrasts. Repeating the identical sign-flip permutation test on the model ensemble’s own per-emotion shifts, and correcting all 66 (source × category × emotion) tests together under Benjamini–Hochberg FDR, gives **7 of 66** significant at $q < 0.05$ (Table 5.14). One of the seven is human: the +4.0pp gain in *guilt* on author-independent texts is significant once tested at this combined scale ($p_{\text{BH}} = 0.041$), where it had not survived correction when tested against the human-only family alone. The other six are model effects.

Interpretation. The human result is the only survivor among the 33 human tests, and it is one specific movement, already noted above as one of the largest descriptive shifts (guilt, author-independent), rather than the general “self-conscious emotions gain, sadness and anger lose” pattern visible in the heatmap; the other 32 human cells, including the numerically larger sadness and fear movements on unambiguous texts, do not survive correction. The six significant model effects form a coherent pattern rather

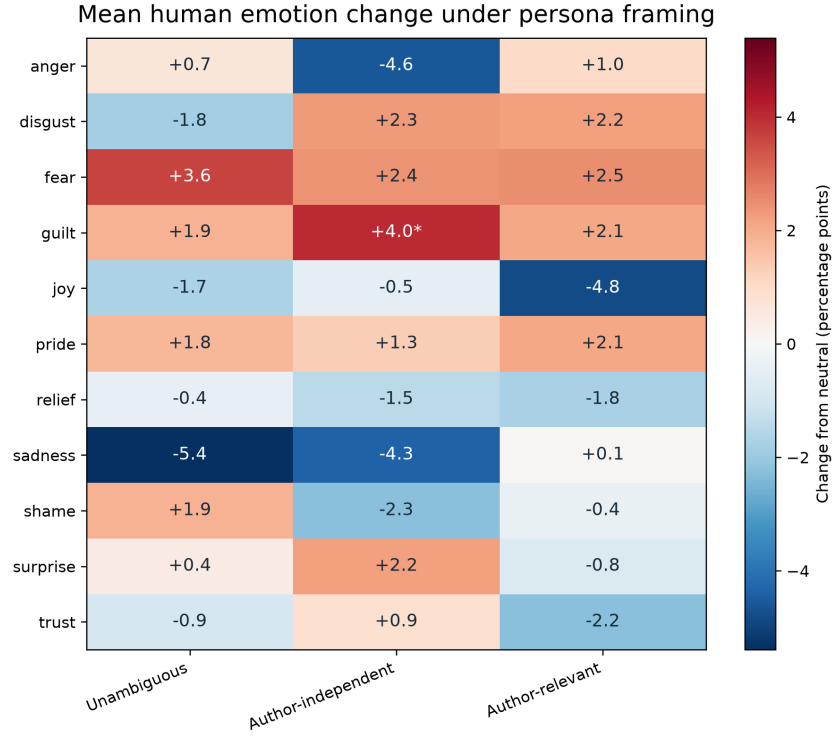


Figure 5.7: Mean signed change in probability mass per emotion and category (human, text-level, percentage points).

than scattered noise: on both author-independent and author-relevant texts the ensemble moves away from *joy* and toward a more specific positive emotion (*pride* on the former, *relief* on the latter), and on unambiguous texts it moves toward *pride* and away from *disgust*. The model ensemble therefore responds to a persona by substituting a more specific emotion for a generic positive one, more than it introduces threat-related emotions. The result is no longer reported as a uniform negative finding; it is a small, mostly model-side, directionally coherent effect superimposed on a human per-emotion pattern that remains largely unconfirmed at this sample size.

Rates at which an emotion appears post-framing that was absent beforehand rise with category for both sources, but are far higher for humans (70.0%, 81.5%, 84.0%) than for the model ensemble (20.3%, 26.3%, 43.4%). Disappearance rates follow the same ordering (humans 52.5%, 66.5%, 73.0%; models 15.7%, 23.1%, 37.3%). What is observed here is a change in the *support*, the set of emotions receiving any vote at all, not a

Table 5.14: Per-emotion shifts significant under BH-FDR at $q < 0.05$ (of 66 tests).

Source	Category	Emotion (mean change)	p_{BH}
Human	Author-Independent	guilt +4.0pp	0.041
LLM	Unambiguous	disgust -0.7pp	0.027
LLM	Unambiguous	pride +1.7pp	0.009
LLM	Author-Independent	joy -1.9pp	0.012
LLM	Author-Independent	pride +1.9pp	0.009
LLM	Author-Relevant	joy -5.1pp	0.009
LLM	Author-Relevant	relief +3.6pp	0.012

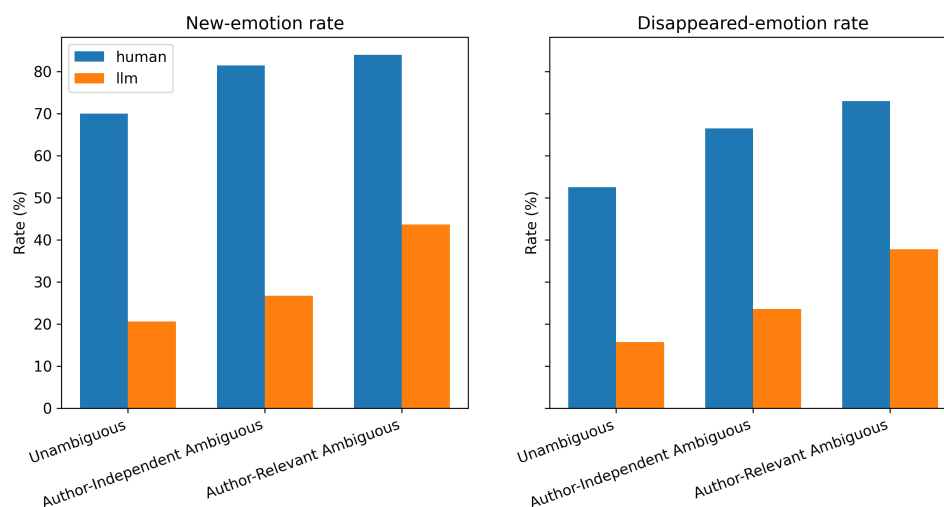


Figure 5.8: Rates at which emotions appear or disappear between conditions, by source and category.

transition tied to any specific rater or model; because the human and persona conditions were rated by disjoint people, no individual vote is ever observed to change. This gap is largely structural rather than behavioural: with only three votes, one human vote landing on a previously unvoted emotion is enough to change the support, whereas a seven-model ensemble requires the same event across a larger denominator, so support detection is itself a function of sample size as well as of the underlying vote distribution.

5.14 A Traceable Strong-Shift and Stable-Condition Example

Text 2105 describes passing exams, with the original emotion word removed from the stimulus. Its neutral human votes are *pride*, *joy*, *joy*. Persona A describes a student who had struggled throughout the year and barely passed; four independent human votes are

sadness, relief, relief, relief. The two votes have disjoint emotion supports, yielding TV distance 1.0 (a 100% distributional shift). Persona B describes a high-performing student who expected to pass; its votes are *pride, joy, joy*, exactly the neutral distribution, giving TV distance 0. This juxtaposition is one illustrative case selected by its observed shift pattern, not evidence that the persona independently caused a particular individual to change their answer. Source: `data/human_annotations.csv` (text 2105), with persona wording in `data/sampled_texts.csv`.

In the zero-shot, numbered-list configuration, the OpenAI model returns *pride* under neutral framing, *relief* under persona A and *pride* under persona B. Its recorded explanations refer, respectively, to accomplishment, relief after a difficult year and success consistent with high expectations (`data/experiment_results_all.csv`, text 2105). These responses show that the explanation field was collected and that this specific model/configuration qualitatively resembles the human contrast; they do not validate the explanations as a mechanism or represent every model/configuration.

5.15 Does Framing Concentrate or Disperse Interpretation?

Magnitude and dispersion are distinct questions. A distribution can shift substantially while landing on a single confident new answer, or spread thinner without moving far. The two are empirically almost unrelated here, in either source, at the text level: for humans $\rho = 0.092$, $p = 0.113$; for the model ensemble $\rho = -0.048$, $p = 0.406$.

Each source-category cell is tested with a one-sample Wilcoxon signed-rank test of that cell’s per-text mean entropy change against zero (each text contributing exactly one value, avoiding the pseudoreplication a per-comparison test would introduce), with Benjamini-Hochberg correction applied jointly across all six tests.

Table 5.15: Entropy change under framing (text-level, BH-FDR corrected across six tests).

Source	Category	Δ entropy	p_{BH}	Significant
Human	Unambiguous	+0.198	0.0002	yes
Human	Author-Independent	+0.171	0.0002	yes
Human	Author-Relevant	+0.161	0.0030	yes
Model	Unambiguous	+0.028	0.226	no
Model	Author-Independent	+0.012	0.230	no
Model	Author-Relevant	+0.021	0.350	no

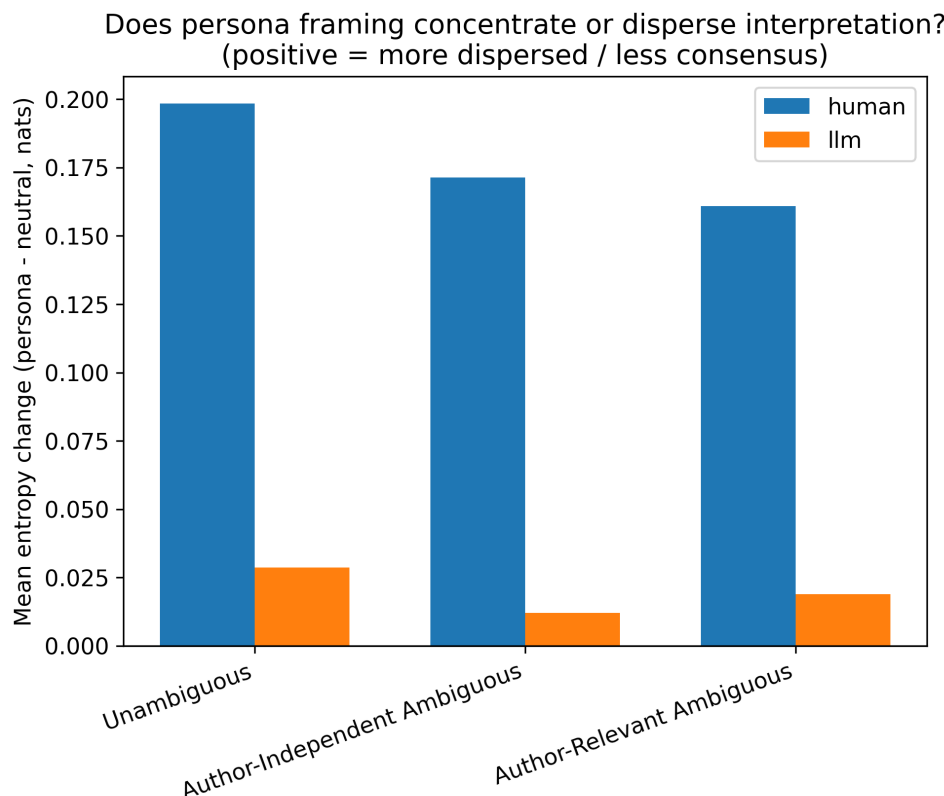


Figure 5.9: Mean entropy change from neutral to persona framing, by source and category. Positive values indicate greater dispersion.

Interpretation. Humans become reliably *less* internally agreed under persona framing, in all three categories. The model ensemble shows no detectable dispersion effect in any category: it moves its probability mass (Section 5.9) without a detected change in how agreed it is with itself. These are two separate sets of tests, one significant throughout and one not, and a significant result alongside a non-significant one does not by itself establish that the two sources differ from each other statistically; no test of that difference directly was performed. Stated cautiously, the pattern is that human dispersion increases were detected and model dispersion increases were not, which is at least suggestive of an asymmetry no magnitude-only measure would reveal, though it stops short of a demonstrated human–model contrast.

A caveat parallels the H4 count-standardisation check (Section 5.12.2): every neutral human cell rests on exactly three votes, whereas persona cells rest on three to seven, and plug-in entropy rises mechanically with the number of votes, so part of the raw

human entropy change above is a vote-count artefact rather than framing. Standardising both conditions to three votes each (`results/count_standardisation_summary.csv`) reduces the mean human entropy change from 0.177 to 0.135 (95% CI [0.089, 0.181]), roughly a quarter smaller, but it remains clearly positive; the model change remains indistinguishable from zero after standardisation as well (0.013, CI [−0.003, 0.029]).

5.16 How Often Does Framing Change Nothing?

The persona changed **nothing at all** (TV exactly zero) in **8.5%** of human comparisons and **40.2%** of model comparisons. Median TV is 60.0% for humans and 14.3% for the ensemble.

Interpretation. The model ensemble’s mean shift of 18.3% does not arise from every cell moving moderately; it arises from a large minority not moving at all and the remainder moving considerably more. This also explains the undefined cosines in Section 5.12.3: a zero delta has no direction. Part of the human–model asymmetry is again structural, since a seven-way vote has more available states than a three-way vote.

5.17 The “Unchanged” Category Conceals Movement

A conventional dominant-label comparison records a change only when the winning emotion flips. Broken into the five outcomes defined in Section 4.4.3:

Table 5.16: Dominant-set outcomes. These set relationships differ from changes in an alphabetically selected single winner.

Outcome	Human	Model ensemble
Identical set	34.3%	78.9%
Single emotion split into a tie	17.0%	2.6%
Tie resolved to a single emotion	8.2%	3.4%
Partial overlap of two ties	7.0%	0.4%
Fully disjoint top sets	33.5%	14.8%

Interpretation. Human top sets are fully disjoint in 33.5% of comparisons and identical in 34.3%; the other 32.2% involve a tie forming, resolving, or partially overlapping. For the model ensemble the corresponding figures are 14.8%, 78.9%, and 6.3%. These are *set* relationships, not the changed/unchanged decisions of the selected-label method.

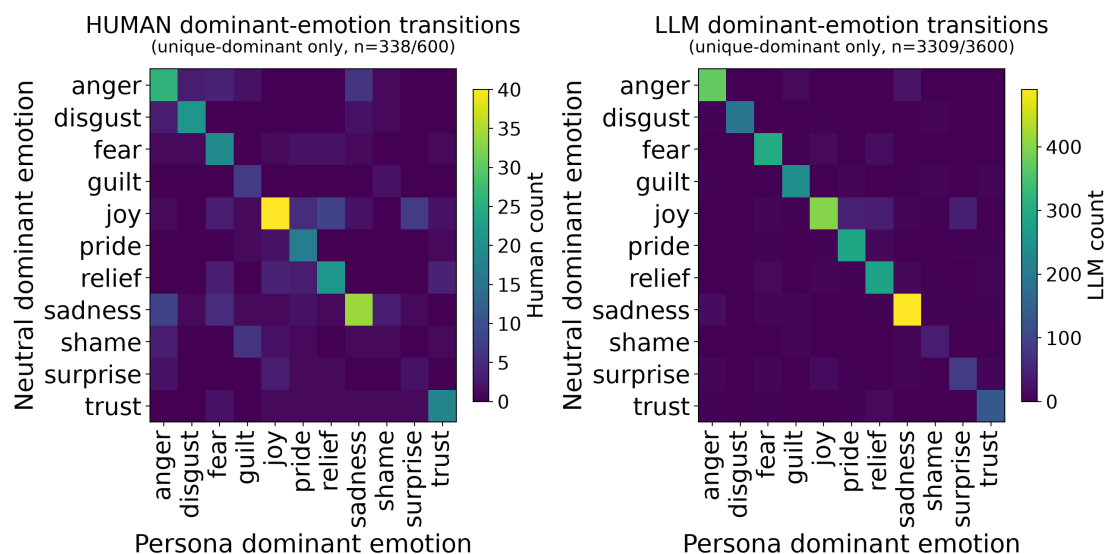


Figure 5.10: Dominant-emotion transition matrices for human and model judgments, restricted to comparisons where both sides have a unique dominant emotion.

That method records 52.8% of human and 18.4% of model comparisons as changed (Section 5.6); an overlapping top set can still select a different winner.

5.18 Prompt Structure Has No Measurable Effect

A repeated-measures Friedman test across the six prompt configurations (text-level, $n = 300$) gives $\chi^2 = 7.298$, $p = 0.199$. The same test on the selected-label change rate rather than the shift gives $\chi^2 = 1.709$, $p = 0.888$; neither representation detects a prompt-configuration effect.

Interpretation. Prompt design, zero-, one- or few-shot, crossed with numbered or inline label presentation, does not measurably change the size of the model framing effect in this dataset. Non-rejection does not establish that prompt configurations are equivalent. Since no numbered hypothesis concerning prompt structure exists in the approved proposal, it is reported as a supplementary observation and must not be cited as testing a formal hypothesis. Its practical value is as a robustness check: the H2 effect is detectable when the six configurations are combined, and descriptive effects are present across configurations; a configuration-specific equivalence claim is not supported.

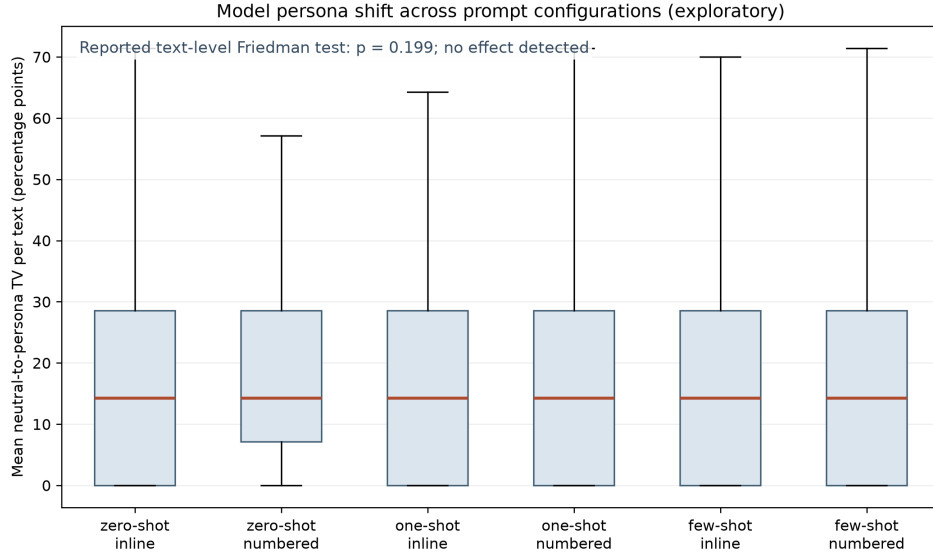


Figure 5.11: Per-text shift magnitude across the six prompt configurations.

5.19 Supplementary Check: Human Rating Agreement

The proposal requested a second check on whether human readers agree. Fleiss’ κ asks: *how much more often do readers agree than we would expect given how common each emotion label is?* A value of zero corresponds to agreement expected from those label frequencies; a higher value means more agreement. It is a supporting description, not another hypothesis or the main reason to use emotion vectors. The statistic is computed for the 754 of 900 text-by-framing cells that each have exactly three recorded human ratings, separately within each sampled category and framing. The neutral cells all have three ratings; some persona cells have four to seven and are omitted from this particular statistic. Table 5.17 gives both kappa and the number of included texts in parentheses (source: `results/human_interannotator_agreement_verified.csv`).

Table 5.17: Human Fleiss’ κ by sampled category and framing; included cells in parentheses.

Category	Neutral	Persona A	Persona B
Unambiguous	0.465 (100)	0.270 (78)	0.350 (76)
Author-Independent Ambiguous	0.344 (100)	0.229 (72)	0.250 (79)
Author-Relevant Ambiguous	0.294 (100)	0.177 (75)	0.159 (74)

Neutral agreement is highest in the unambiguous pool and lowest in the author-relevant pool. The persona columns describe only the exactly-three-vote subset, with different included texts across framings. They therefore neither establish a causal change in agreement nor replace the within-text framing analysis above.

5.20 Summary of Hypothesis Outcomes

Table 5.18: Summary of hypothesis outcomes.

Hypothesis	Key evidence	Verdict
H1	Author-relevant significant both sides ($p_{\text{Holm}} = 0.0066, 0.0112$); unambiguous non-significant; author-independent A only, direct A–B inconclusive; selected-label tests: 1 of 6 significant under reverse-alphabetical tie-break, 0 of 6 under alphabetical	Partially supported
H2	Paired test $p_{\text{Holm}} = 0.0002$ both sides; also significant under the selected-label cross-check; 21.3% of model pairs changed label; generic-author control shows movement for both neutral-to-generic and generic-to-specific steps (not additive components of one total effect)	Supported
H3	Raw: 3 of 3 contrasts Holm-significant. Null-referenced: 0 of 3	Predicted raw ordering observed; framing-specific contrast inconclusive
H4 magnitude	34.86pp difference, surviving count standardisation ($\approx 34.9 \rightarrow \approx 33\text{pp}$); positive text-level covariation (Figure 5.5)	Supported (magnitudes differ)
H4 direction	Mean cosine 0.112, $p = 0.0001$ vs chance pairing	Partially supported
Per-emotion (exploratory)	7 of 66 (source $\times$ category $\times$ emotion) tests significant under BH-FDR: 1 human (guilt), 6 model	Small, mostly model-side effect
Prompt structure	Friedman $p = 0.199$ (vector), $p = 0.888$ (selected-label)	No effect detected (exploratory)

Hypothesis outcomes

Primary tests and key sensitivity results

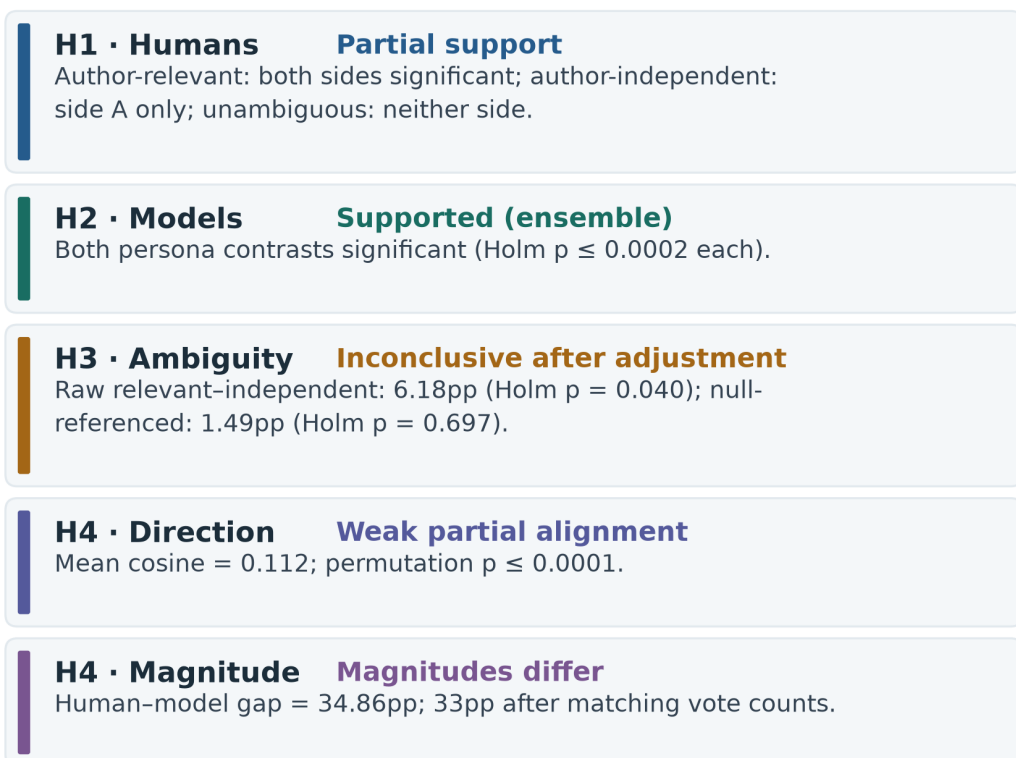


Figure 5.12: Overview of hypothesis outcomes across the analysis.

6 Discussion

This chapter interprets the results against the three research questions, draws out findings that cut across individual hypotheses, states the limitations of the design candidly, and identifies what further work would resolve.

6.1 Interpretation with Respect to the Research Questions

6.1.1 RQ1: Does Perceived Authorship Influence Human Emotion Judgments?

The evidence for a framing effect is category-dependent.

Human judgments shift significantly under author framing for author-relevant ambiguous texts on both persona sides; the tests do not detect an effect for unambiguous controls on either side. That pattern is consistent with H1. Non-rejection in the controls does not establish a zero effect or isolate the persona’s content from other changes introduced by the author cue.

The author-independent category reaches significance under persona A and does not under persona B. This split in threshold decisions cannot by itself show that the personas have different effects. The exploratory direct contrast of their text-level raw TV shifts is 4.44 percentage points, with a 95% bootstrap interval from -2.08 to 11.06 points (Section 5.8). Because this interval includes zero, these data do not establish an A-versus-B asymmetry. The category’s possible response to incidental persona content remains an interpretation to investigate, not a demonstrated result.

A qualification must be attached to the magnitude of these effects, and it is developed in Section 5.11.2. A substantial share of the raw human shift is what one would expect from three-vote sampling noise alone, given each text’s own vote distribution. The effect is detected in the sense that it exceeds its permutation null, but the raw TV magnitude

mixes that detected effect with ordinary sampling noise, so it should not be read as a precise measure of framing’s own contribution to the movement.

A second qualification concerns *how* the effect was found at all. Repeating the identical permutation test on the selected label rather than the vote vector rejects the null in one of the six cases under a reverse-alphabetical tie-break convention, and none of the six under the alphabetical convention used throughout this thesis (only these two conventions were tested; Section 5.8.2). Almost all of H1’s support for RQ1 is therefore visible only under the distributional representation; a study identical in every other respect but analysed by majority label would have detected a statistically significant effect in one of the six cases under the reverse-alphabetical convention, and under this thesis’s own convention, none. That gap is largely an artefact of the representation, though the exact size of the gap depends on an otherwise-arbitrary tie-breaking choice rather than being a fixed property of the data.

6.1.2 RQ2: Does Perceived Authorship Influence LLM Classifications?

The answer is yes, and with less qualification than for RQ1.

The ensemble effect is significant on both persona sides. All seven models have nonzero observed label-change rates, but separate per-model significance tests were not conducted. The exploratory test detects no prompt-configuration effect; it does not establish equivalence across formats. The observed changes across models and formats make the aggregate finding less dependent on any single setting, subject to the single-generation limitation below. This extends the finding of Santurkar et al. 2023 that persona conditioning shifts model outputs, from opinion elicitation to emotion classification, and matches the direction of Zhang et al. 2018 on persona-conditioned dialogue.

The generic-human control sharpens the claim considerably. The neutral-to-generic-author contrast is approximately 8% on both sides (7.98% and 8.01%), while the generic-author-to-specific-persona contrast is 16.2% and 19.3%. These distances show sensitivity both to the presence of an attribution and to its specific content, but they are not additive components of a total effect, so they cannot establish what proportion of the overall shift is due to identity specifically. Without this control, the two could not have been distinguished at all.

Two boundaries on the claim should be stated. First, the significance test detects *co-ordinated* redistribution across models; 61.2% of cells contribute no variance to its null,

and a scenario in which every cell contained exactly one dissenting model would yield $p = 1$ despite universal change. The complementary raw change rate, 21.3% of model pairs altering their label, is reported precisely because the two measures answer different questions. Second, each model produced one generation per condition, so none of this separates a framing response from ordinary stochastic variation in generation. Repeated generations under identical prompts would be required to do that, and they were not collected.

6.1.3 RQ3: Do LLMs Resemble Humans in Direction and Magnitude?

The answer divides cleanly, and the division is the most substantive finding of the thesis.

In direction, models resemble humans weakly but reliably. Mean cosine between human and model delta vectors is 0.112, well above a shuffled-pairing null. Models do not merely shift when framing is applied; they shift in a direction related to the direction the same text moves for human readers. But a cosine of 0.11 is a small association. The resemblance is real and it is partial, which is exactly what H4 predicted. This is a more demanding criterion than the usual label-agreement comparison: it asks not whether the model reaches the same answer as a human, but whether it *responds to context* in the same direction, which is closer to the question posed by ambiguity-aware approaches such as Hong et al. 2025.

In magnitude, models and humans differ substantially. Humans shift more by roughly 34 percentage points, and this survives standardising both sides to an equal number of votes: doing so reduces the estimated gap by only about 1.5 points (roughly 4%), and it does not remove the between-person/within-model design asymmetry discussed below.

There is a further pattern that magnitude alone does not capture, though it should be read cautiously. Under framing, humans become measurably *less internally agreed*: entropy rises significantly in all three categories. The model ensemble shows no detectable dispersion effect in any category. These are separate tests, and a significant result on one side alongside a non-significant result on the other does not by itself establish that humans and models differ from each other statistically; no direct test of that difference was performed. Reported separately rather than as a demonstrated contrast, the pattern is at least suggestive that models redirect their probability mass while human judgments

also become more dispersed, which would be invisible to any analysis reporting shift magnitude alone.

Taken together, the results show sensitivity to author framing, weak directional alignment with human judgments, and smaller observed shifts. Human entropy increases were statistically detected, whereas model entropy increases were not; whether those entropy responses differ between humans and models remains untested.

6.2 Cross-Cutting Findings

6.2.1 What Aggregation Costs

The clearest methodological result of this thesis is quantitative rather than argumentative. A selected-label analysis of these same data would have recorded 38.7% of human comparisons and 41.4% of model comparisons as showing no change, when the underlying distribution demonstrably moved (under the alphabetical tie-break convention; crediting the label method for tracking the complete set of tied top emotions instead lowers these to 25.8% and 38.7% respectively). For the model ensemble, the comparisons such an analysis cannot see outnumber those it detects by more than two to one, and for humans the invisible comparisons are nearly three-quarters as numerous as the detected changes (38.7% invisible against 52.8% detected).

This is not an argument that label-based analyses are wrong for all purposes. It is a demonstration that for *this* question, whether framing shifts interpretation, the choice of representation determines whether the phenomenon is observable at all. The clearest case is the human side: a study designed identically but analysed on majority labels would have detected a statistically significant human framing effect in one of the six H1 tests (reverse-alphabetical tie-break convention) or none (this thesis’s alphabetical convention) rather than three, and would most plausibly have been read as showing little or no human sensitivity to author framing. The model-side selected-label test still detects a significant effect either way (Section 5.9.1), so for the model ensemble the representation choice changes how much of the shift is visible, not whether an effect is detected at all.

This gives a concrete magnitude to a concern raised in principle by Aroyo and Welty 2015 and by Basile 2020, and it is consistent with Pavlick and Kwiatkowski’s 2019 argument that disagreement in subjective annotation is frequently inherent rather than erroneous.

Where that literature argues that aggregation discards signal, the measurement here states how much and of what kind.

The related finding on baseline agreement reinforces the point. Human and model dominant labels agree for only 58.0% of comparisons under neutral framing overall (ranging from 74.2% on unambiguous texts down to about 50% on the two ambiguous categories), but within both the matching and non-matching buckets, full-vector similarity spans nearly the entire possible range. Reporting only the binary match rate compresses a continuous and highly variable quantity into a single number that conceals its own variance, and even that single number shows the two sides starting from a noticeably different reading of the ambiguous texts before any persona is introduced.

6.2.2 Small Samples Make TV Deceptive

A finding that emerged from methodological scrutiny rather than from the original design deserves emphasis, because it generalises beyond this thesis.

Total variation distance between two small vote samples is not zero when no effect is present, and it grows with the dispersion of the underlying distribution. Two independent three-vote draws from the same distribution routinely disagree, and they disagree more when that distribution is spread out. Since categories defined by ambiguity are, by construction, categories defined by dispersion, comparing raw TV across such categories confounds the quantity of interest with the measurement's own noise floor.

Referencing each text against its own conditional sampling null showed that expected null TV is itself ordered by category, from roughly 43% to 55%. The excess above that conditional-null expectation is on the order of two to six percentage points, not forty-five to sixty. Any study comparing distributional distance across groups that differ in baseline dispersion faces this problem, and it is not visible without constructing the null explicitly.

6.3 Limitations

The following limitations bound what these results can support. They are stated here rather than distributed through the results chapter so that they can be weighed together.

Vote counts are small and unequal between sides. Human vectors rest on three raters per condition and model vectors on up to seven. This drives several human–model asymmetries reported in Chapter 5 (tie frequency, new-emotion rates, zero-shift rates, dominant-set stability) which are structural rather than behavioural. Count standardisation (Section 5.12.2) addresses the magnitude comparison specifically, but not these descriptive asymmetries.

The human comparison is between-person; the model comparison is within-model. Different people rated each framing condition, whereas the same model answered both. Part of the measured human shift is between-rater variability with no model counterpart. This is intrinsic to the design and is not remedied by matching vote counts. It is the principal caveat on the H4 magnitude result.

H3’s categories were partly defined by the property H3 tests. Texts were assigned to the author-relevant group using an LLM judge’s own sensitivity to whether author framing flips the reading. H3 is therefore a transfer comparison: whether a grouping derived from that judge’s sensitivity predicts effects in newly collected data, and not an independent discovery that author-relevance is the operative principle. For the model-side ordering specifically (Section 3.4), this is a sharper limitation: the category was defined by an LLM’s persona sensitivity, and the model ensemble’s framing effect is then reported as ordered by that same category, so part of that ordering is true by construction rather than an independent finding.

The category ordering is not separable from baseline dispersion. H3’s contrasts are significant in raw TV and not significant once referenced against each text’s sampling null. This is not because the null-referenced intervals are wider; they are, if anything, narrower in absolute width than the raw contrasts’ intervals (Section 5.11.2). Rather, the estimated excess above the conditional null shrinks far more than the interval does, so the same interval width now straddles zero. This does not demonstrate equivalence between categories, but it does show that this design cannot attribute the raw ordering to framing rather than to baseline dispersion at 100 texts per category.

Model stochasticity is unmeasured. One generation per condition means the model figures are single draws, not stable per-model expectations.

No generic-human condition was collected for humans. The decomposition separating author presence from persona content is available for models only, so the equivalent human question remains open.

Per-emotion effects are underpowered. Only 7 of the 66 per-emotion tests survive FDR correction (Section 5.13), and 32 of the 33 human tests do not, including the numerically largest movements in the heatmap. The broader descriptive pattern may be real, but this study establishes only the one human effect and the six model effects that survived correction, not the pattern as a whole.

Cross-text respondent dependence cannot be modelled, and within-text independence is not fully guaranteed either. No single collection form repeated a text on both of its conditions, which is a property of the form design and is verifiable from the assignment records. But respondent identifiers were not retained, and nothing links one participant's responses across different forms, so if the same person completed more than one form, that person could in principle have rated both conditions of the same text without this being detectable in the retained data. The same absence of participant tracking means cross-text correlation (one person's answers on different texts being related) also cannot be ruled out or quantified; each collection form carried on the order of thirty items, so if forms and participants corresponded one to one this would bound the possible correlation structure, but that correspondence is itself not verifiable from what was retained. The within-text permutation null also requires comparable respondent groups under the null; disjoint form assignments narrow this risk but do not, on their own, establish that exchangeability assumption.

6.4 Future Directions

Five extensions follow directly from the limitations above.

Collect human annotations at matched vote counts. Raising human raters per condition to seven would eliminate the principal structural asymmetry and allow the descriptive comparisons, tie rates, new-emotion rates, dispersion, to be interpreted behaviourally rather than arithmetically.

Add the generic-human condition to the human study. This would permit the same neutral-to-generic-author versus generic-to-specific-persona comparison already run on the model side, so that the two separately measured distances could be compared directly for humans instead of assumed from the model pattern.

Sample repeated generations per model. Several generations per prompt would separate framing response from generation variance and convert the model figures from single draws into estimates with their own uncertainty.

Define ambiguity categories independently of the tested property. Assigning texts to categories on criteria not derived from an LLM judge’s own framing sensitivity would convert H3 from a transfer comparison into a direct test.

Retain respondent identifiers. Anonymised respondent identifiers linked to text and condition would allow a participant-aware analysis and remove the one dependence structure this study cannot quantify.

A further direction is suggested by the entropy finding rather than by a limitation. That a significant human dispersion increase was detected while no model dispersion increase was, though this apparent contrast was not directly tested, is, to the author’s knowledge, not something the existing literature on persona prompting has examined. It connects to work modelling annotators individually rather than in aggregate [Davani, Díaz, and Prabhakaran 2022; Uma et al. 2021]: if human interpretive uncertainty is itself structured, a model that reproduces answers without reproducing uncertainty is capturing only part of the behaviour. Whether models can be made to exhibit human-like interpretive uncertainty, rather than merely human-like answers, appears to be an open and tractable question.

7 Conclusion

This thesis asked whether perceived author identity measurably changes how emotion is interpreted in ambiguous text, for human readers and for large language models, and whether the two respond in comparable ways. It answered the question by manipulating author attribution experimentally while holding textual content constant. A first pass selected one emotion per condition and tested the hypotheses with that label. The second pass retained all eleven emotion proportions, tested the hypotheses again, and showed which changes the first pass had missed.

7.1 Summary of Findings

Author framing shifts human emotion judgments for author-relevant ambiguous texts, on both persona sides, and shows no statistically detectable effect for unambiguous control texts. The latter result is consistent with H1 but cannot confirm a zero effect or establish that the manipulation acted only on author information. In the author-independent category one persona contrast is significant and the other is not; a direct comparison does not establish different effect sizes.

The model ensemble is sensitive to author framing on both persona sides. All seven models have nonzero observed label-change rates, while separate per-model significance tests were not run. No prompt-configuration effect was detected, so equivalent effects across formats cannot be claimed. A previously unanalysed generic-human control separates this into two steps under matched models: adding an unspecified author line alone moves classifications by about 8% on both sides, and the further step to a specific persona moves them by a further 16–19%. These are two separately measured distances, not additive components of one total effect.

The predicted ordering across ambiguity categories is present in raw shift magnitude and all three pairwise contrasts survive correction. However, referencing each text against its own conditional sampling null shows that expected shift under no effect is itself strongly

ordered by category, and that the excess above that conditional-null expectation is small enough that none of the contrasts remains significant. The raw ordering agrees with H3’s prediction, but evidence for a greater *framing-specific* effect in the author-relevant category is inconclusive at this sample size.

On human–model alignment, the two components of H4 separate cleanly. Models track the *direction* of human shifts better than chance but only weakly, which is what H4’s word “partially” predicts. Human shifts are substantially larger in *magnitude*, by roughly 34 percentage points, and this survives standardising both sides to equal vote counts. Beyond magnitude, a further pattern emerged that the original hypotheses did not anticipate, reported cautiously since the two sides were not tested against each other directly: under framing, human judgments become measurably less internally agreed, while the model ensemble shows no statistically detectable dispersion effect in any category. Taken separately rather than as a demonstrated contrast, these results are at least suggestive that models redirect probability mass while human group judgments also become more dispersed.

Two results are reported as negative or near-negative findings. Almost no individual per-emotion shift survives multiple-comparison correction: of 66 tests spanning both sources, only one human effect (a gain in guilt on author-independent texts) and six model effects reach significance, and the model effects form a coherent pattern of moving away from joy toward more specific positive emotions (pride, relief) rather than toward threat-related ones. No effect of prompt configuration on framing-effect magnitude was detected in the exploratory comparison. A further, unplanned result is that most of H1’s support depends on the choice of representation: the identical permutation test applied to the selected label rather than the vote vector rejects the null in one of its six cases under a reverse-alphabetical tie-break convention, and none under the alphabetical convention used throughout this thesis (only these two otherwise-arbitrary conventions were tested), so a majority-label version of this same study would have detected a statistically significant human framing effect in one of those six tests at best, or none under this thesis’s own convention.

7.2 Contribution

The substantive contribution is evidence that perceived authorship is a real and measurable influence on emotion interpretation in both humans and language models, together

with a control condition isolating persona content from author presence, and a direct comparison of the two sides under one design.

The methodological contribution is arguably the more portable one. Representing judgments as distributions rather than labels is not a presentational preference; it determines whether the phenomenon is observable. A selected-label analysis of these same data would have recorded 38.7% of human and 41.4% of model comparisons as showing no change when the distribution demonstrably moved (under the alphabetical tie-break convention; crediting the label method for tracking the complete set of tied top emotions instead lowers these to 25.8% and 38.7% respectively): for models, more than twice the comparisons it would have detected, and for humans nearly three-quarters as many as it would have detected. More directly still, that same label-based treatment finds *no* significant human framing effect under the alphabetical tie-break convention used as the primary analysis, and one of six under the reverse-alphabetical convention also tested, where the distributional treatment finds three of six. Any study asking whether some manipulation shifts subjective interpretation faces this exposure, and the size of the blind spot is not obvious without measuring it.

A second methodological result has similar reach. Distributional distance between small vote samples is not zero under a null of no effect, and it scales with the dispersion of the underlying distribution. Comparing such distances across groups that differ systematically in dispersion, as any ambiguity-defined grouping does, confounds the effect of interest with the measurement’s own noise floor. An explicit per-item sampling null makes this visible, and in this study it changed what could be concluded from H3.

7.3 Closing Remarks

The practice this thesis set out to question, collapsing multiple human judgments into one gold label and treating the residue as noise [Basile 2020; Aroyo and Welty 2015], is convenient, entrenched, and for a substantial fraction of texts simply not well defined. Among the sampled texts (a sample stratified toward ambiguous categories, not a corpus-wide rate), one text in five has no emotion commanding a majority of readers and 13.3% have no unique most-popular answer at all. For the texts with tied top choices, a pipeline reporting a single label depends on a tie-break; for texts with a unique plurality but no majority, that label still conceals the disagreement.

What the distributional view shows is that the disagreement being discarded is not simply noise. Its overall magnitude and direction shift detectably when readers are told something about who wrote the text, in humans and in models alike, and that shift is weakly but measurably shared between the two (Section 5.12); this is a claim about the aggregate shift, not about every specific per-emotion pattern or category ordering examined along the way, several of which remain inconclusive at this sample size. Emotion recognition systems evaluated only against majority labels are being scored on a target that discards the variation this thesis has shown to be both structured and responsive to context.

Whether models should reproduce human interpretive uncertainty, as opposed to merely producing human-like answers, is a design question rather than an empirical one. But it cannot even be posed while the evaluation target is a single label, and the results reported here suggest the two differ in ways that matter.

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